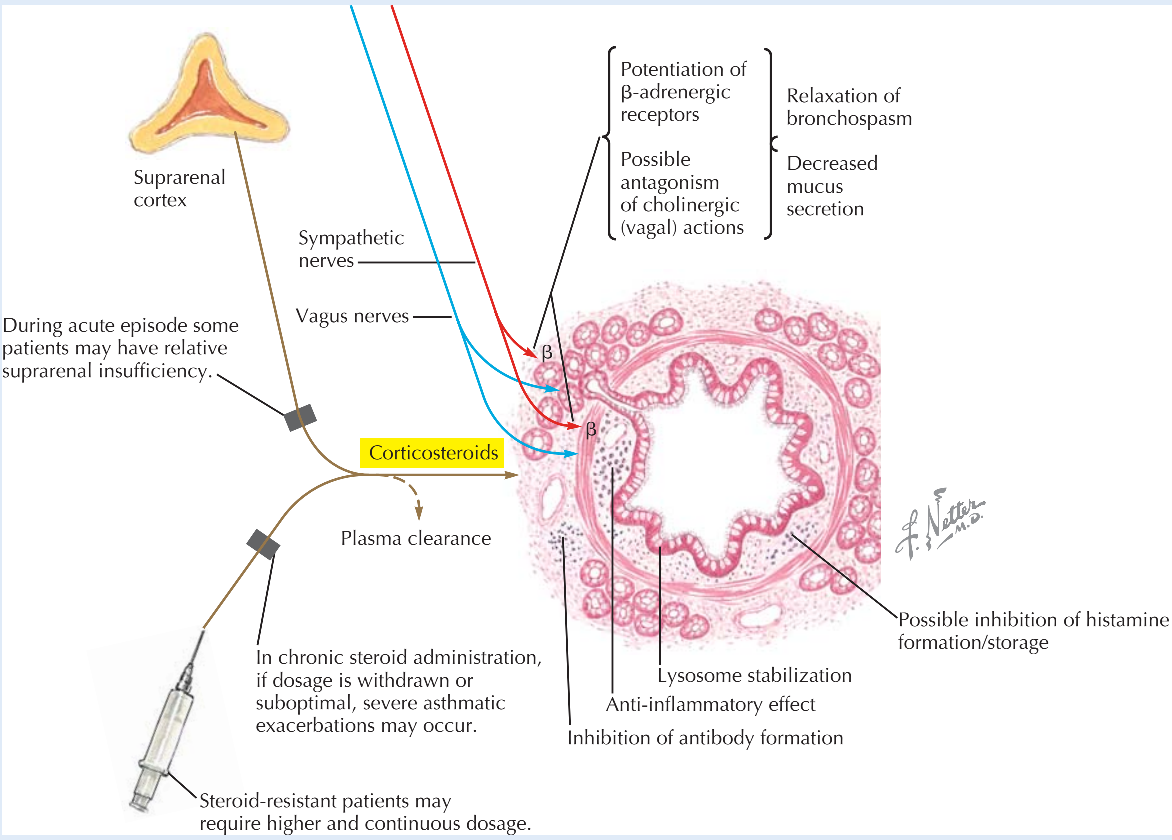
ACTIONS OF ADRENAL ANDROGENS

ACTH stimulates the zona reticularis of the adrenal cortex to synthesize and release androgens, although additional factors are likely involved. The adrenal androgens, dehydroepiandrosterone (DHEA) and androstenedione, have notable effects in females, because the adrenal gland is normally the only significant source of androgens in women. Effects of adrenal androgens in females include development of pubic hair, hypertrophy of sebaceous glands (acne), stimulation of libido, and possibly, inhibition of osteoporosis. They have only a minor role in men, in whom testosterone produced by the testes is the major androgen. However, in both sexes, adrenal androgen production rises in mid childhood, and thus the hormones may contribute to early pubertal changes in boys as well as girls (Chapter 31). **Adrenarche** is the maturation of the adrenal cortex, which occurs between 6 and 10 years of age and results in the three distinct zones of the cortex, and subsequently, production of adrenal androgens. Adrenal androgen levels reach a peak at about 20 years of age and decline through adulthood.

CLINICAL CORRELATE
Glucocorticoids as Anti-inflammatory and Immunosuppressive Drugs

Based on their anti-inflammatory and immunosuppressive properties, cortisol and various synthetic glucocorticoids (for example, dexamethasone) are used therapeutically to suppress serious inflammation and allergic reactions, autoimmune responses, and transplant rejection. However, systemic treatment with steroids is avoided when possible, due to a wide array of potentially serious side effects, including immunosuppression, muscle wasting, osteoporosis, hyperglycemia, neural excitability and associated psychiatric effects, and, ultimately, adrenal insufficiency (due to negative feedback on the HPA axis). When an anti-inflammatory steroid is used systemically, the patient is gradually weaned from the drug, to prevent sudden withdrawal before endogenous cortisol production can resume.



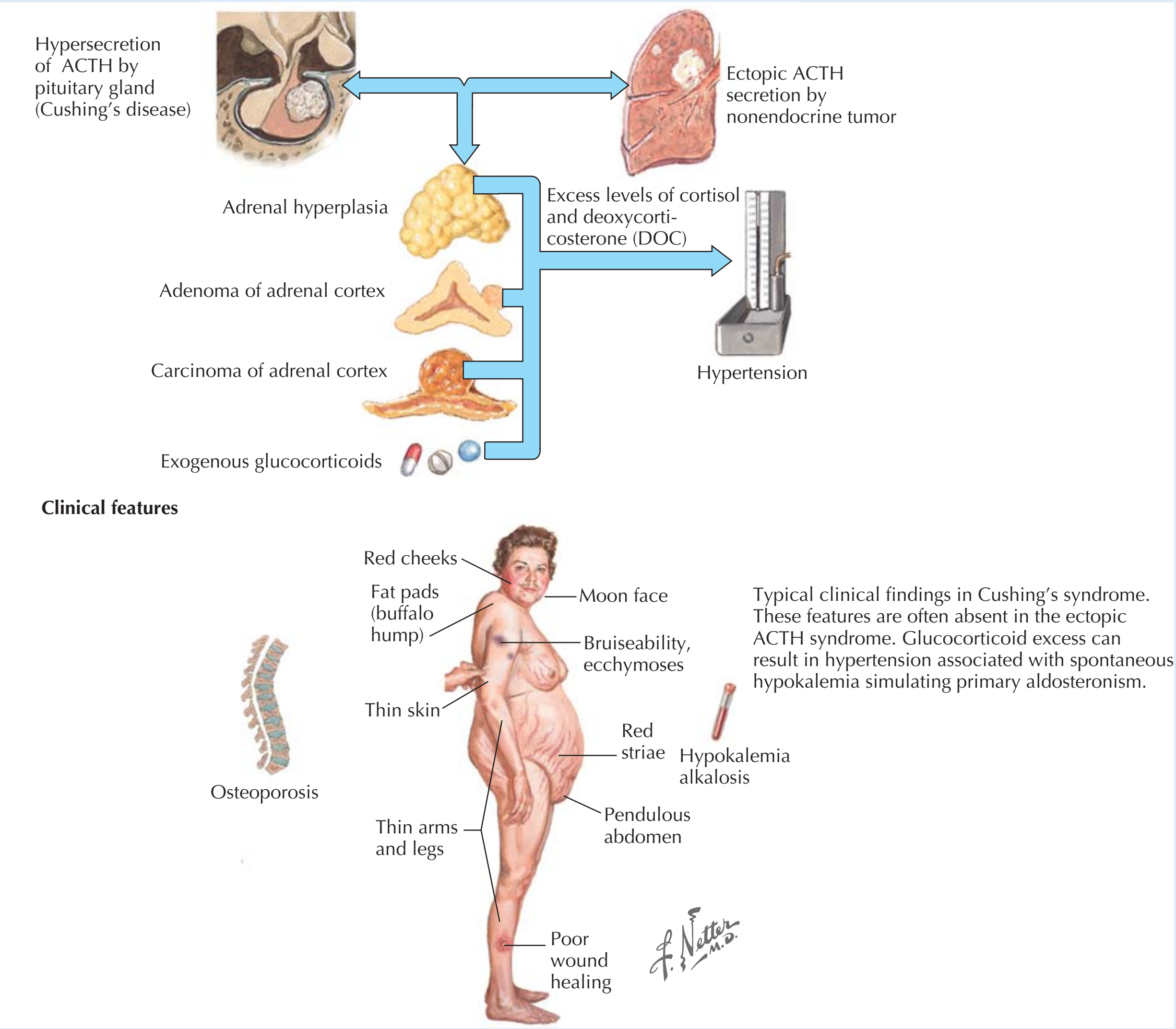
Corticosteroid Actions in Bronchial Asthma Anti-inflammatory effects are the basis for use of inhaled corticosteroids as preventive medication in patients who suffer frequent or severe asthma attacks.

CLINICAL CORRELATE
Cushing's Syndrome

Cushing's syndrome is an endocrine disease caused by high levels of blood cortisol. Its etiology may be due to a pituitary or ectopic ACTH-secreting tumor, hyperplasia or tumor of the adrenal gland, or use of glucocorticoid drugs. The symptoms and signs of Cushing's syndrome, which might be predicted based on the effects of adrenal steroids, include:

- centripetal obesity, accompanied by muscle wasting in the extremities
- "moon face" (a rounding of the face)
- thinning and bruising of the skin
- skin hyperpigmentation (if ACTH is elevated; see earlier in the chapter)
- hypertension
- hyperglycemia and insulin resistance
- osteoporosis

The dexamethasone suppression test may be useful in diagnosing the cause of hypercortisolism. Although secretion of ACTH by the pituitary (and thus, adrenal cortisol production by the adrenal gland) is normally suppressed by administration of dexamethasone (a glucocorticoid drug), pituitary ACTH-secreting tumors have reduced sensitivity to such inhibition. When hypercortisolism is the result of an adrenal tumor, cortisol production is unaffected by exogenous glucocorticoid (dexamethasone) administration. Once the cause of glucocorticoid excess is identified and appropriately addressed (often by surgery), treatment with replacement steroids is usually necessary.



Causes of Cushing's Syndrome Cushing's syndrome may result from a variety of causes, all of which result in elevated plasma glucocorticoid level. ACTH, adrenocorticotropic hormone.

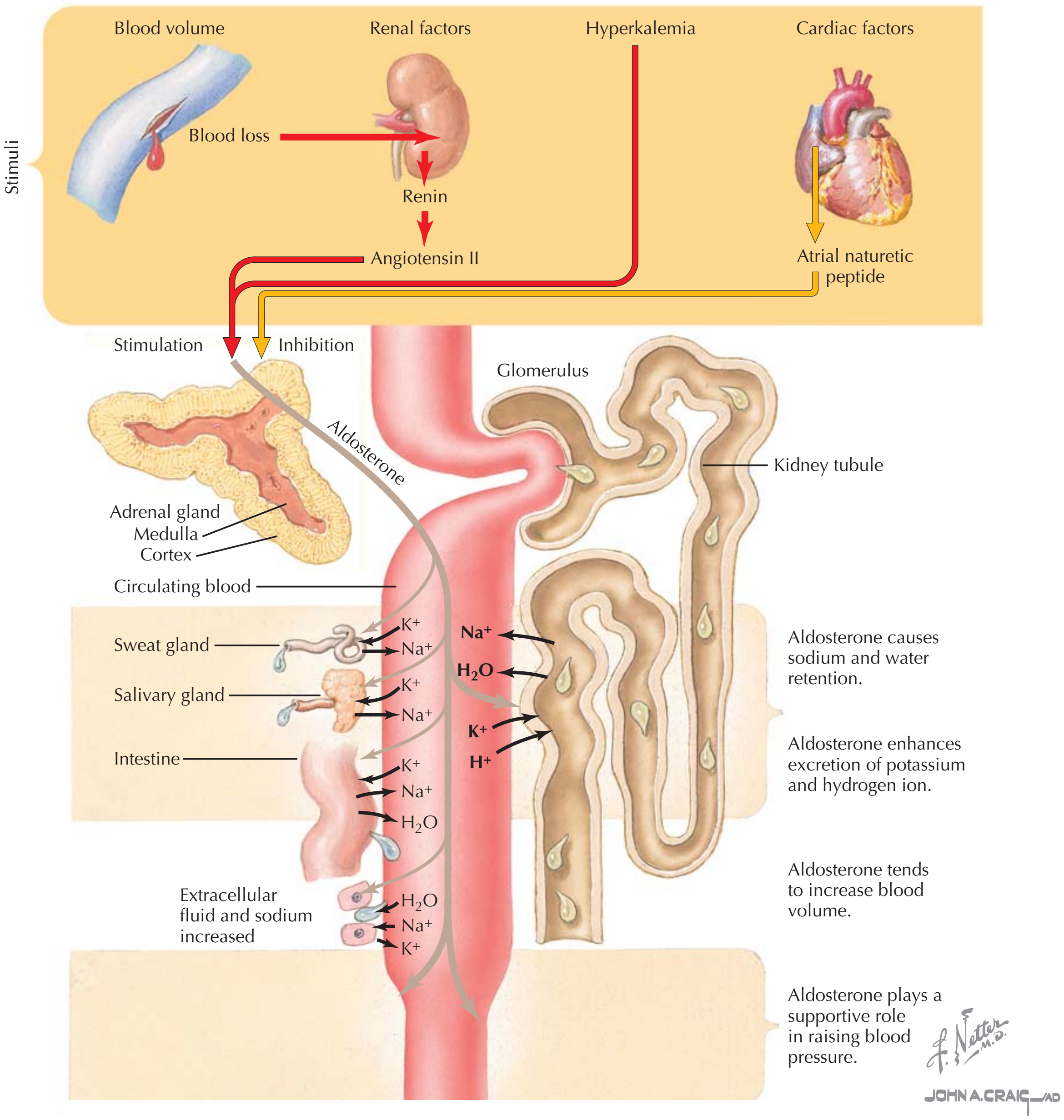


Figure 28.5 **Actions of Aldosterone** The steroid hormone aldosterone has the important functions of regulating extracellular fluid volume and K^+ levels. Synthesis and release of aldosterone is promoted by angiotensin II and hyperkalemia and is inhibited by atrial natriuretic peptide. Its action results in water and Na^+ retention and K^+ and H^+ excretion by the kidney; it has similar effects on the intestine, sweat glands, and salivary glands.

REGULATION AND ACTIONS OF ALDOSTERONE

The mineralocorticoid aldosterone is a regulator of **extracellular fluid volume** and **K^+ homeostasis** (Fig. 28.5). Its actions are accomplished mainly by effects on the collecting ducts and late distal tubules of the kidney, where it stimulates:

- **Na^+ reabsorption,** and as a result, water retention and **expansion of extracellular fluid volume;** in excess, hypertension results.
- **K^+ excretion;** excess aldosterone will cause hypokalemia.
- **H^+ excretion;** in excess, aldosterone causes metabolic alkalosis.

In addition to the renal tubules, aldosterone also affects Na^+ and K^+ handling by the intestines, salivary glands, and sweat glands.

The primary stimuli for secretion of aldosterone by cells of the adrenal zona glomerulosa are (see Fig. 28.5):

- **Hyperkalemia** (elevated plasma K^+), which directly acts on the adrenal gland to increase aldosterone secretion; aldosterone reduces plasma K^+ through its renal actions.
- **Angiotensin II,** which stimulates aldosterone secretion. Reduction of blood volume results in renin release by the kidney, which cleaves the plasma protein angiotensinogen to angiotensin I, which is subsequently cleaved by angiotensin-converting enzyme (ACE) to form angiotensin II (see Section 5, "Renal Physiology"). Aldosterone, by promoting Na^+ and water retention, increases blood volume.
- **Atrial natriuretic peptide (ANP),** which inhibits aldosterone secretion. ANP is released by cardiac myocytes when blood volume is elevated, for example, in congestive heart failure. Inhibition of aldosterone secretion results in reduction of blood volume.

While ACTH is required for synthesis of aldosterone, the above listed factors play a more important regulatory role in its synthesis and release. Release of aldosterone, like release of cortisol and adrenal androgens, is greatest in the early morning hours. The physiology of aldosterone is discussed in greater detail in Section 5, "Renal Physiology."

THE ADRENAL MEDULLA

In addition to the following discussion, physiological effects of adrenal catecholamines are discussed throughout the various chapters of this book in relation to the physiology of various target systems and in the specific context of the autonomic nervous system as a whole in Chapter 7.

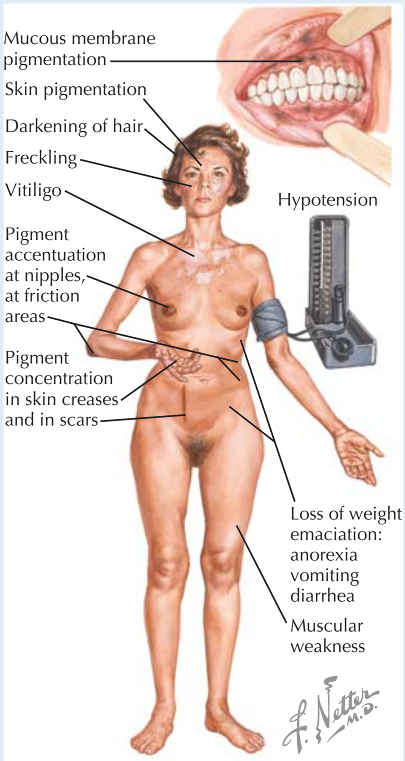
The adrenal medulla, the central portion of the adrenal gland, is functionally distinct from the cortex, acting as a postganglionic effector of the sympathetic nervous system. Its **chromaffin cells** synthesize and release catecholamines into the bloodstream in response to sympathetic nervous system activation. Whereas sympathetic postganglionic nerves mainly release **norepinephrine,** 70% to 80% of the catecholamine release from chromaffin cells is **epinephrine.** As is the case for postganglionic neurons, a number of sympathetic cotransmitters, including **neuropeptide Y,** are also synthesized and released. Release of adrenal catecholamines results in a spectrum of physiological actions throughout the body, with effects differing somewhat in degree compared with effects of norepinephrine released from sympathetic nerves, owing to the differences in binding affinity of epinephrine and norepinephrine at subtypes of adrenergic receptors (Fig. 28.6).

CLINICAL CORRELATE
Addison's Disease

Addison's disease is an uncommon endocrine disease characterized by failure of steroid synthesis by the adrenal gland, due to various causes including autoimmune or tubercular destruction of the gland and rare genetic disorders. As is the case for Cushing's syndrome, effects of deficiency of adrenal steroids might be predicted based on the normal actions of the hormones. Symptoms and signs of adrenal insufficiency include:

- poor stress tolerance
- hypoglycemia
- fatigue and weight loss
- hyperpigmentation of skin (due to ACTH elevation; see earlier in the chapter)
- low blood pressure
- salt appetite

An Addisonian crisis, caused by a severe deficiency of adrenal hormones, is a medical emergency, sometimes marking the onset of the disease. Vomiting and diarrhea, low blood pressure, fainting, loss of consciousness, convulsions, and hypoglycemia are among the signs and symptoms. Addisonian patients require long-term glucocorticoid replacement therapy, and sometimes mineralocorticoid treatment as well.



Chronic Primary Adrenocortical Insufficiency (Addison's Disease) In primary adrenocortical insufficiency, adrenocortical steroid production is low, due to adrenal atrophy, tubercular destruction of the adrenal, or other causes, and ACTH level is elevated (due to lack of negative feedback). Symptoms reflect deficiency of corticosteroids but also reflect excess of ACTH, which causes pigmentation due to its sequence homology with α-melanocyte-stimulating hormone (MSH). ACTH, adrenocorticotropic hormone.

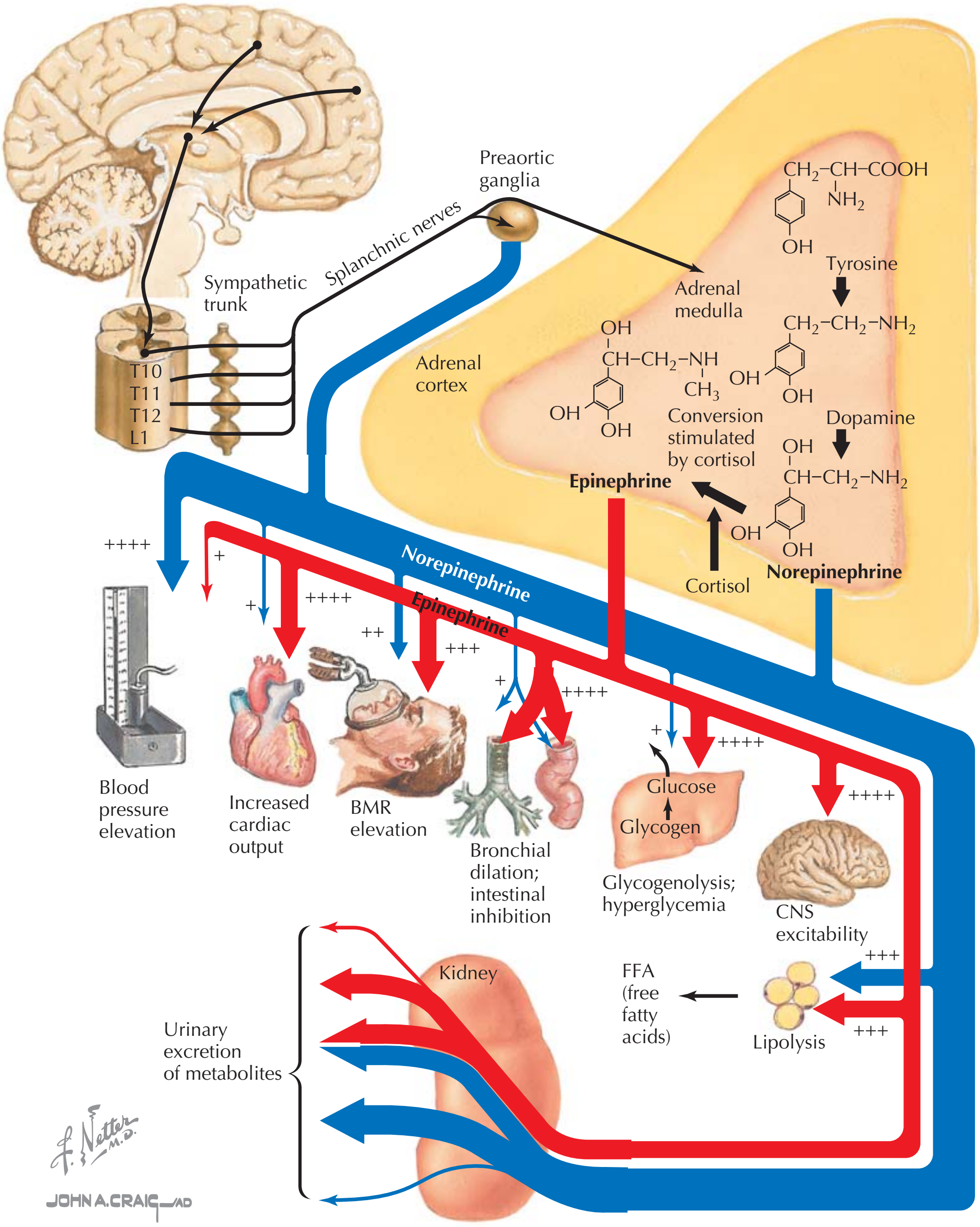


Figure 28.6 **Function of the Adrenal Medulla** The adrenal medulla releases epinephrine (80%) and norepinephrine (20%) into the bloodstream during activation of the sympathetic nervous system. The effects of epinephrine and norepinephrine, including relative magnitude, are illustrated for various sites. BMR, basal metabolic rate.

CHAPTER 29
The Endocrine Pancreas

The pancreas has key roles in both gastrointestinal function and modulation of blood glucose levels. Because the products of the exocrine pancreas (enzymes and buffers) allow digestion and absorption of carbohydrates into the blood, and the products of the endocrine pancreas (insulin, glucagon, and somatostatin) regulate the blood glucose levels, the pancreas can be viewed as a total processing unit for glucose entry into the body and the cells.

STRUCTURE OF THE PANCREAS

As noted in Section 6, the exocrine pancreas performs crucial digestive and buffering functions. The exocrine pancreas makes up about 99% of the functional cells, with the remainder serving the endocrine function (Fig. 29.1). The primary endocrine portion is composed of the islet of Langerhans cells, which are made up of three cell types:

- α-Cells produce **glucagon,** which mobilizes glucose stores *into the blood.*
- β-Cells produce **insulin,** which stimulates glucose transport *into the cells.*
- δ-Cells produce **somatostatin,** which inhibits both insulin and glucagon secretion.

In the islets, the cells are organized with the α-cells surrounding the β-cells. There are many more insulin-producing β-cells than α and δ cells. There is a fourth cell type (F-cell) within the islets, which produce pancreatic polypeptide, which inhibits exocrine secretions (buffers and enzymes).

There are also important paracrine (cell-to-cell) interactions within the islets. Because the blood forms a capillary network in the islets, insulin (from β-cells) is carried to the glucagon-producing α-cells in a paracrine manner. Insulin can inhibit its own secretion, as well as glucagon secretion, whereas somatostatin inhibits both insulin and glucagon. This allows paracrine modulation of the pancreatic response to blood glucose levels.

> Because both insulin and the **C-peptide fragments** are secreted when contents of the insulin-containing granules are released, the amount of C-peptide in the blood is a reflection of insulin production. Clinically, C-peptide is used to determine endogenous insulin secretion in diabetic patients receiving insulin injections.

SYNTHESIS OF INSULIN, GLUCAGON, AND SOMATOSTATIN

Insulin

The major control of blood glucose levels occurs through the production and secretion and action of **insulin,** which is a 51 amino acid peptide hormone formed in the β-cells from a prohormone that contains three peptides: the A and B chains of the active insulin molecule and the connecting "C"-peptide (Fig. 29.2). In the endoplasmic reticulum, disulfide bridges are formed between the A and B chains of the proinsulin, which are still connected by the C-peptide chain. In the Golgi apparatus, the C-peptide is cleaved from the proinsulin, forming active insulin. Both the insulin and C-peptide are packaged in granules for secretion into blood.

Glucose (GLUT) Transporters

Secretion of insulin is stimulated directly by blood glucose levels, as well as by gut peptides. The glucose enters cells through a variety of facilitated glucose transporters (GLUT). Although many GLUT transporters have been identified (more than 10), the function of several of the newly discovered isoforms is not clear. Of the main transporters:

- **GLUT1** is in all adult cell membranes and is responsible for allowing enough glucose into cells to maintain cellular respiration and viability. It is also found in high concentration in the membranes of the blood-brain barrier, and the transporters are under positive control by circulating glucose (increased glucose increases GLUT1 transporters). **SGLUT1** are secondary active transporters, which carry glucose with sodium, and these are found in *apical* membranes of renal proximal tubules, choroid plexus, and small intestine along with **SGLUT2.**
- **GLUT2** is located in the membranes of the small intestine (basolateral side), brain, liver, and pancreas, and these transporters facilitate easy entry of glucose to those tissues. SGLUT2 are secondary active transporters,

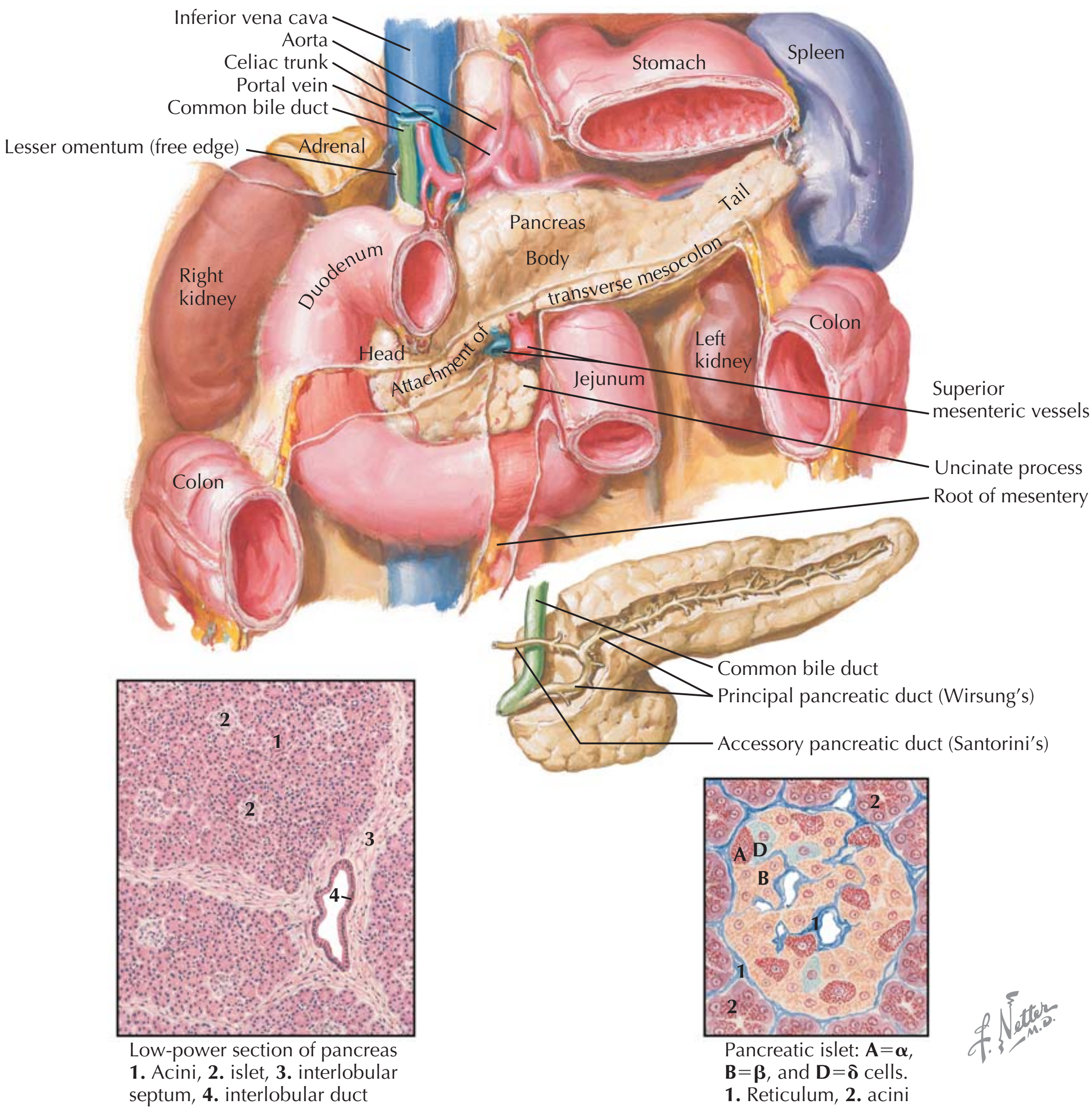


Figure 29.1 Structure of the Endocrine Pancreas The pancreas is a key component of the gastrointestinal (GI) tract because of its exocrine function; it also provides the primary control of blood glucose because of its production of endocrine hormones. The vast majority of the pancreas (~99%) is composed of acinar cells, which produce and secrete the buffers and enzymes through ducts into the GI tract (exocrine function) *(micrograph on left)*. The endocrine pancreas is composed of cells that form the islets of Langerhans. The cells of the islets produce insulin (β-cells), glucagon (α-cells), and somatostatin (δ-cells).

which carry glucose with sodium, and these are found in *apical* membranes of renal proximal tubules, choroid plexus, and small intestine along with SGLUT1.

- **GLUT3** is mainly in neurons and placenta.
- **GLUT4** is expressed in skeletal and cardiac muscle and adipose tissue and is the **insulin-stimulated glucose transporter.**
- **GLUT5** is the fructose transporter.

Mechanism of Insulin Secretion

When blood glucose increases (see Fig. 29.3*A*):

- Glucose enters the pancreatic β-cells via **GLUT2** transporters.
- Intracellular glucose metabolism increases adenosine triphosphate (ATP), which inhibits K^+ efflux. This depolarizes the β-cells and opens voltage-gated Ca^{2+} channels.
- The Ca^{2+} influx stimulates secretion of the insulin and C-peptide into the blood.

In addition to blood glucose, the gut-derived **incretins** (glucose insulinotropic peptide [GIP] and glucagon-like peptide-1 [GLP-1]) and increased blood amino acids, fatty acids, and

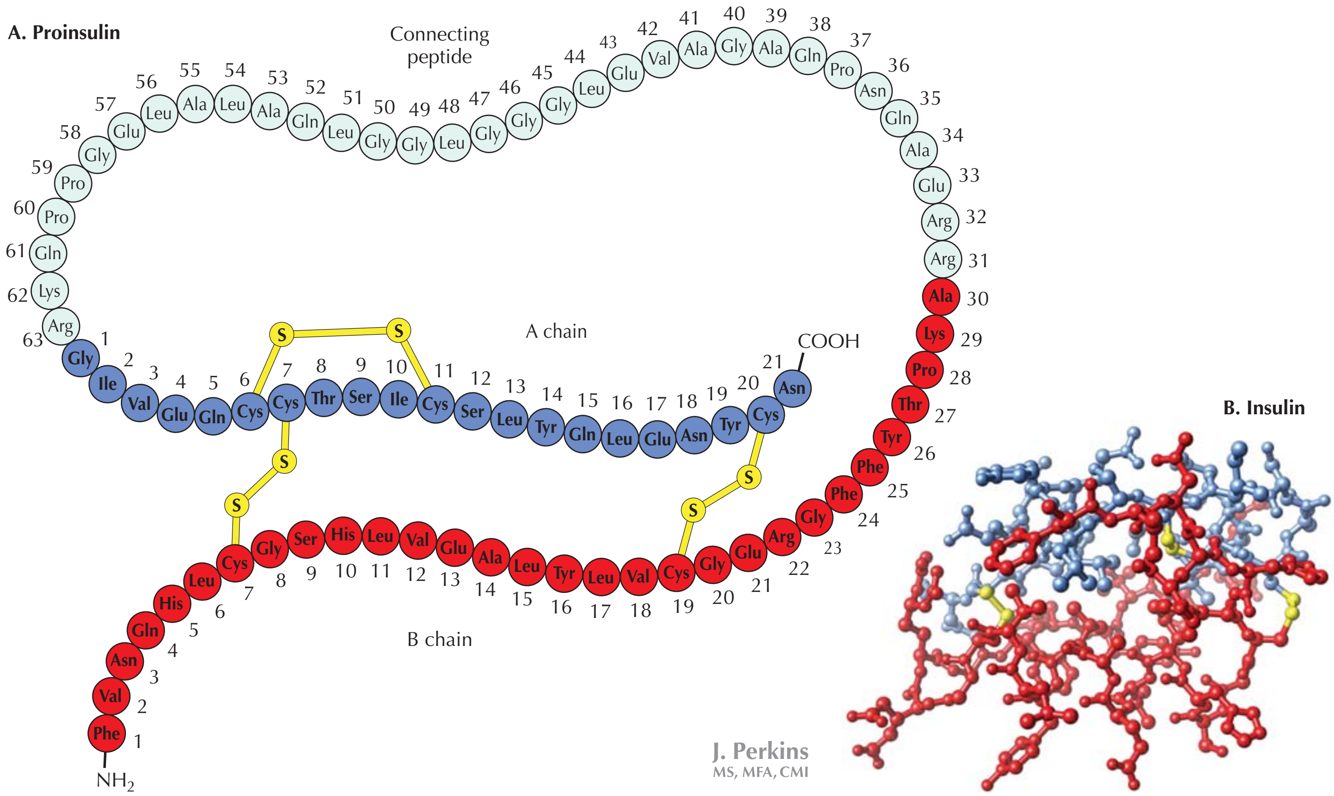


Figure 29.2 **Insulin Structure** *A* illustrates the proinsulin molecule, which is composed of an A *(blue)* and B *(red)* chain of insulin, connected by two disulfide bridges *(yellow).* In the endoplasmic reticulum, the insulin chains are attached to the connecting "C"-peptide, which is cleaved in the Golgi apparatus to yield the insulin and C-peptide and then packaged in secretory granules; the three-dimensional structure of active insulin is illustrated in *B.*

acetylcholine also promote insulin secretion (Fig. 29.3*B*). A common theme is that all of these elements reflect the fed state as a stimulus for elevating insulin. Lastly, glucagon can also stimulate insulin secretion by increasing intracellular Ca^{2+} (via phospholipase C). This modulates the hyperglycemic effects of glucagon.

Insulin secretion is suppressed when blood glucose is low, when the sympathetic nervous system is stimulated (elevated norepinephrine and epinephrine), and when local somatostatin is elevated.

Glucagon

Glucagon is a 29 amino acid peptide hormone produced as a prohormone in the α-cells of the islets of Langerhans. As with insulin, intracellular processing results in packaging of active glucagon molecules in dense core granules.

Glucagon acts to mobilize glucose into blood (actions opposite of insulin); the secretion of glucagon is stimulated primarily by low blood glucose. Insulin inhibits glucagon secretion in a paracrine manner. Thus, modulation of glucagon is observed with ingestion of amino acids, which increases both insulin and glucagon. Glucagon secretion is also inhibited by high glucose and fatty acids in the blood.

Somatostatin

Although not well understood, **somatostatin** plays a lesser role in control of blood glucose. It is a 14 amino acid peptide produced in the δ-cells of the islets and acts in a paracrine manner to suppress *both* insulin and glucagon. This adds one more level of control in modulating blood glucose levels. The somatostatin produced in the pancreas is the same hormone that is produced in the brain and gut, and as in these other tissues, it acts locally on adjacent cells.

Pancreatic polypeptide is another endocrine hormone secreted from F-cells in the pancreatic islets. Although the function is not clear, it is released in response to meals high in protein, and it appears to decrease secretion of gastric and intestinal enzymes, modulating the effects of cholecystokinin (see Chapter 23).

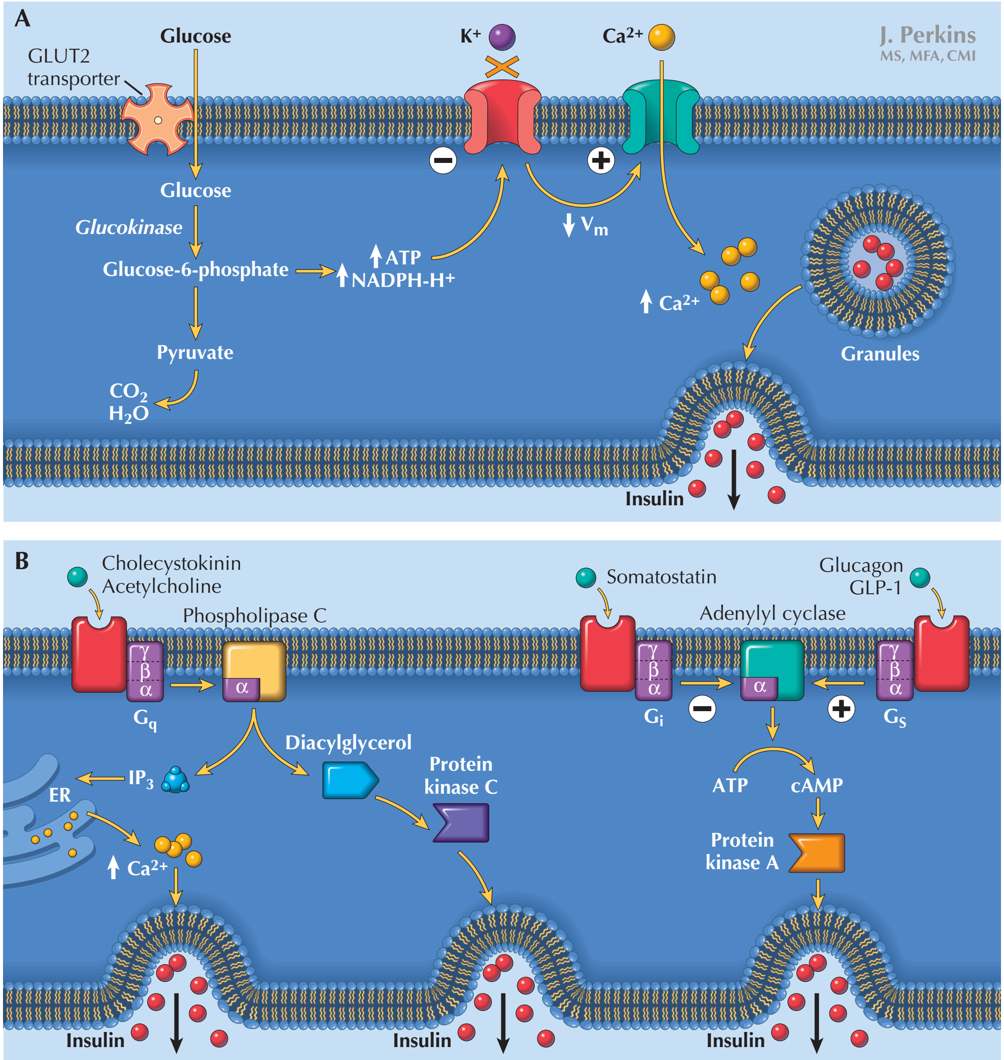


Figure 29.3 **Insulin Synthesis and Release** Glucose is the most important factor regulating insulin synthesis and secretion, although gastrointestinal peptides and local glucagon and somatostatin also contribute to modulation of release. *A* depicts the action of glucose to increase Ca^{2+} influx, which stimulates insulin secretion. *B* illustrates the receptor-mediated *stimulation* of insulin secretion by local glucagon, gut peptides (CCK and GLP-1), and ACh, and *inhibition* of insulin by local somatostatin. ACh, acetylcholine; CCK, cholecystokinin; GLP, glucagon-like peptide.

ACTIONS OF PANCREATIC HORMONES

The primary role of the pancreatic hormones is to maintain an appropriate basal level of glucose in the blood, which in humans is 70 to 90 mg%. To accomplish this, the hormones sequester glucose into cells when there is an influx during feeding—the hypoglycemic effect of insulin—and mobilize glucose out of cells during fasting—the hyperglycemic effect of glucagon. This overall balance is accomplished by the integration of glucose metabolic activity primarily within liver, muscle, and adipose tissues, all orchestrated by the pancreatic hormones.

Insulin

As discussed earlier, elevation of blood glucose level will stimulate insulin secretion. Overall, insulin promotes entry of glucose into cells, synthesis of glycogen stores, and reduced lipolysis (Fig. 29.4). These anabolic functions ensure that nutrients are stored and thus available to tissues between meals (fasting).

Whereas the insulin receptor is expressed on most tissues, the receptor concentration is high in liver, muscle, and adipose tissue. The insulin receptor consists of two extracellular

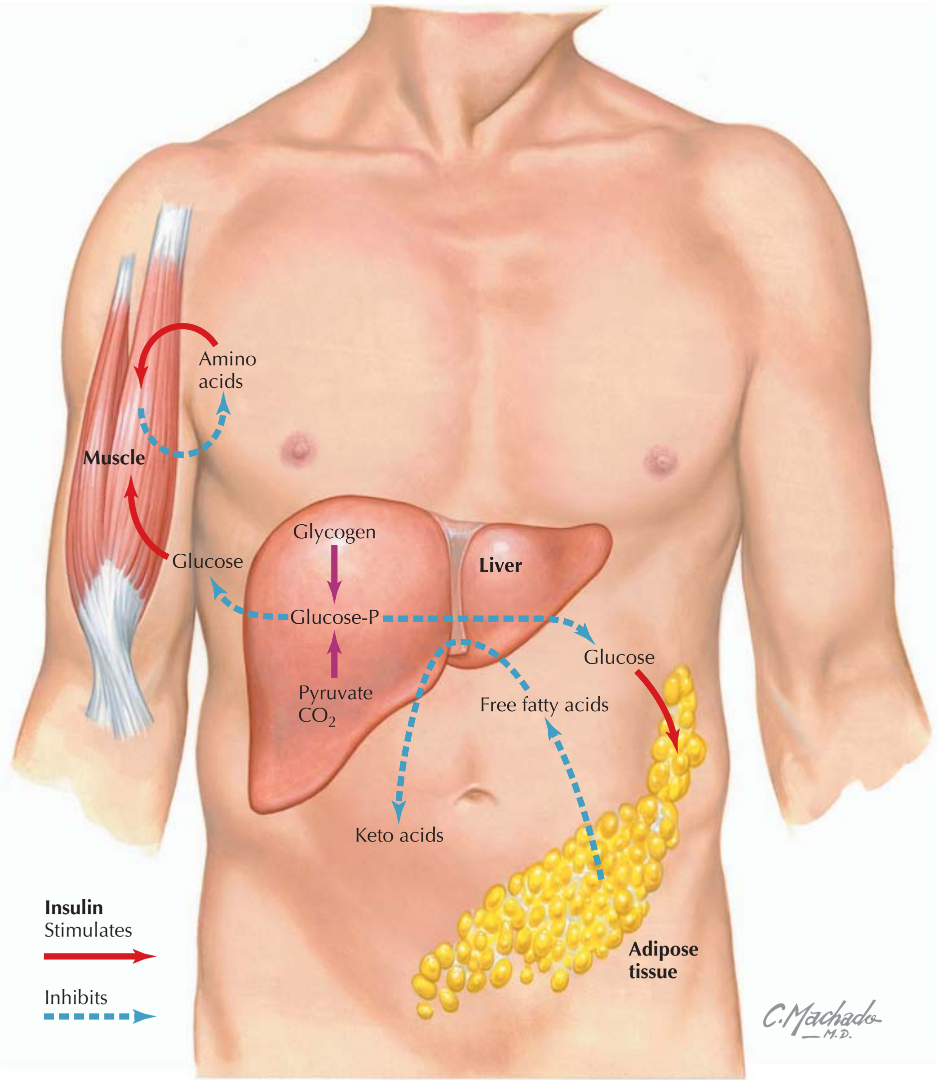


Figure 29.4 **Actions of Insulin** Insulin is considered a "fuel storage" hormone, and therefore insulin promotes the storage of glucose (as glycogen) and fatty acids (as triglycerides [TG] in adipose tissue). Insulin stimulates the uptake of glucose into cells via GLUT4 transporters, and the glucose is used or stored as glycogen. The major glycogen stores are in muscle and liver. Insulin also stimulates fat synthesis and inhibition of lipolysis in adipose tissue, which maintains stores of TGs and reduces keto acid production. Lastly, insulin stimulates uptake of amino acids into skeletal muscle and storage as protein. The overall result is that insulin decreases plasma glucose, fatty acids, and keto acids.

α-subunits and two β-subunits (transmembrane). Insulin binds to the α-subunits and stimulates phosphorylation of the β-subunits, as well as phosphorylation of other intracellular proteins. Thus, the ability of insulin to phosphorylate various intracellular proteins causes its hypoglycemic effect by:

- Increasing GLUT4 transporters in membranes, allowing efficient glucose entry into cells.
- Increasing glycogen production from excess glucose, to facilitate storage.
- Inhibiting glycogenolysis, to maintain the glycogen storage.
- Inhibiting gluconeogenesis, to prevent production and release of glucose back into the blood.

All of these factors rapidly reduce the blood glucose concentration when insulin is present. Furthermore, insulin also has effects on lipid metabolism by:

- Inhibiting hormone-sensitive lipase in adipose tissue, which decreases lipolysis and reduces the amount of circulating free fatty acids.
- Inhibiting oxidation of fatty acids (especially in the liver), which decreases keto acid formation.

Once the receptor phosphorylation occurs, the entire receptor complex is internalized and then is degraded, stored, or reinserted in the membrane. Insulin also has the ability to down-regulate its receptors through increased degradation and decreased synthesis.

> Insulin is a critical metabolic hormone; without it most cells cannot take up glucose. The exception is in brain, liver, and exercising muscle, which have adequate normal glucose uptake due to the presence of GLUT2 transporters (and up-regulation of GLUT4 in exercising muscle). Until the discovery of insulin, type 1 diabetes was a fatal disease.

Glucagon

When blood glucose levels fall during periods of fasting, glucagon is released. In opposition to insulin, glucagon promotes the use of the cellular energy stores, releasing glucose into the blood (Fig. 29.5). The glucagon receptor is a G protein–coupled receptor and is a member of the secretin-glucagon family of receptors, sharing homology with secretin and GIP receptors. Unlike insulin receptors, glucagon receptors are mainly on the cell membranes in the liver and kidney. The liver has stores of glycogen as well as the ability to oxidize fatty acids at a high rate, and glucagon stimulates gluconeogenesis in both organs.

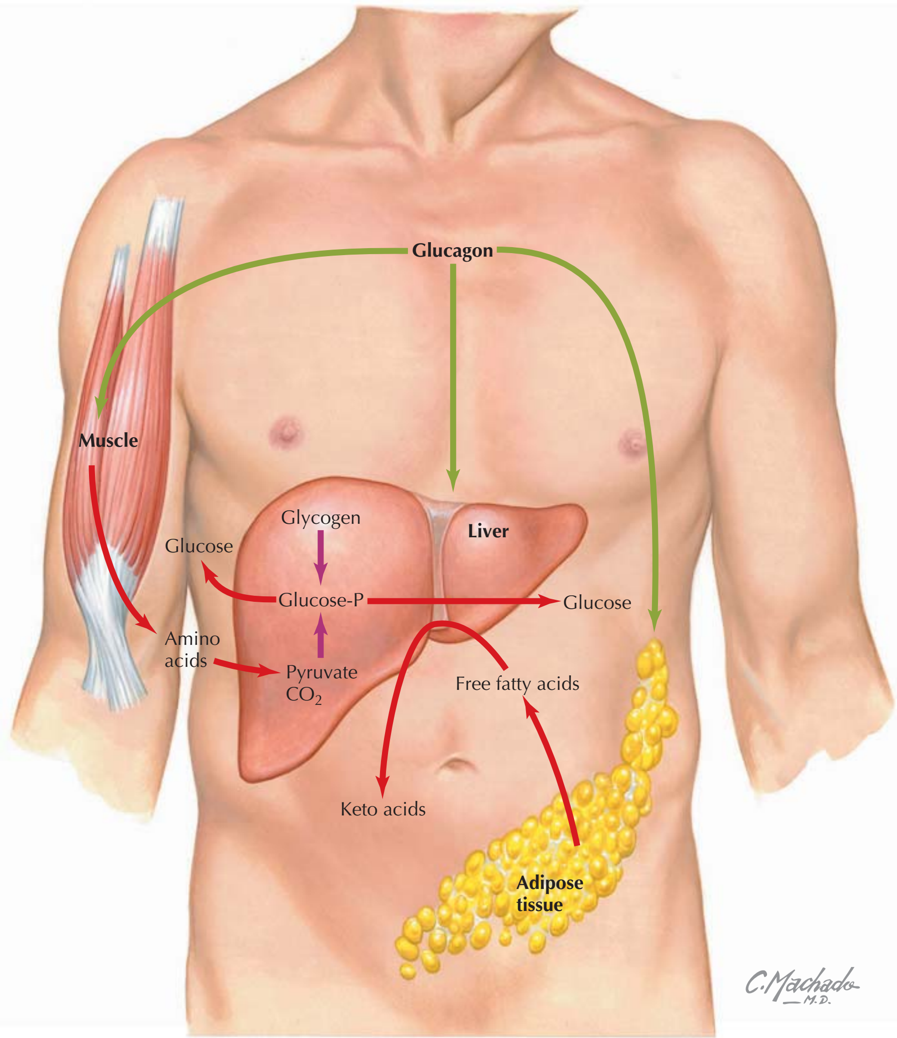


Figure 29.5 Actions of Glucagon Glucagon is considered the "fuel mobilization" hormone, because it breaks down glycogen, protein, and lipids, releasing glucose, amino acids, fatty acids, and keto acids into the blood to serve metabolic demand. Glucagon is stimulated by low blood glucose and promotes glycogenolysis and gluconeogenesis in the liver, increasing blood glucose. Glucagon also stimulates lipolysis and release of fatty acids from adipose tissue, which are oxidized to keto acids in the liver. Lastly, in muscle, glucagon inhibits protein synthesis providing amino acids for conversion to glucose via gluconeogenesis in the liver.

Binding of glucagon with its receptor stimulates cAMP, which then activates kinases that phosphorylate various enzymes. These enzymes contribute to the hyperglycemic effect of glucagon by:

- inhibiting glycolysis
- increasing gluconeogenesis
- increasing glycogenolysis, breaking down glycogen stores, and releasing the glucose into the blood

These factors produce a rapid increase in blood glucose levels. In addition, glucagon has effects on lipid metabolism by increasing β-oxidation of fatty acids.

In normal fasting between meals, there is a balance between glucagon mobilizing glucose and insulin replenishing cellular glucose. However, with prolonged fasting, the glucagon effect predominates, and following depletion of glycogen stores, there are high rates of gluconeogenesis and fatty acid oxidation. The increased oxidation of fatty acids to acetyl coenzyme A (acetyl-CoA) produces **ketone bodies** (acetoacetate and hydroxybutyrate), which can contribute to the acid load of the body (Chapter 20).

Both insulin and glucagon are stimulated by certain **amino acids.** Although this may seem counterintuitive, if a meal is high in protein, but low in carbohydrates, insulin will be increased, stimulating both amino acid and glucose uptake into cells—this lowers the plasma glucose concentrations and would produce hypoglycemia, *if not for the increase in glucagon*. Thus, glucagon is modulating the insulin response.

In addition, although both insulin and glucagon reduce plasma amino acid concentrations by increasing uptake into the hepatocytes, their objectives are different. Insulin promotes peptide synthesis; glucagon uses the amino acids for gluconeogenesis. Thus, amino acids contribute to the mobilization of glucose in the fasted state.

Somatostatin

Somatostatin is secreted in response to all ingested nutrients, as well as glucagon, and is inhibited by insulin. Somatostatin's role in the pancreas is to suppress insulin and glucagon secretion. The apparent disconnection between these actions suggests that somatostatin acts to modulate the secretion of the major hormones.

CLINICAL CORRELATE
Diabetes Mellitus

Diabetes mellitus is a disease of impaired insulin function, resulting in hyperglycemia. There are two classifications of diabetes, which relate to their etiology. **Type 1 diabetes** (T1D) is what used to be called juvenile diabetes or insulin-dependent diabetes mellitus (IDDM); **type 2 diabetes** (T2D) was known as adult-onset diabetes or non–insulin-dependent diabetes mellitus (NIDDM).

T1D is caused by progressive destruction of the pancreatic β-cells by autoimmune attack, eventually resulting in minimal release of insulin and hyperglycemia. The autoimmune (T-lymphocyte) destruction may be caused by viral infection, but the exact causes are not well understood. The onset of hyperglycemia occurs rapidly and usually becomes evident before the person reaches 20 years old. The lack of insulin creates two immediate problems: (1) the glucose cannot enter the cells efficiently, and (2) the lack of glucose in adipose cells decreases fat synthesis and increases fat breakdown, releasing fatty acids into the blood—they get oxidized to ketone bodies by the liver and can result in ketoacidosis (see Chapter 20). Treatment is to stabilize the acid–base balance and start insulin injections. Careful monitoring of blood glucose levels and insulin injections are needed throughout life.

In contrast, **T2D** appears to be a form of **insulin resistance,** resulting from a reduction in the insulin receptors on target tissues. If insulin is not present (T1D), or the receptors are reduced (T2D), the GLUT4 receptors will not be increased at the target cells, and glucose will not efficiently enter the cells and will be elevated in the blood. Although insulin resistance is hereditary, the down-regulation of receptors is also exacerbated with obesity. There are several ethnic groups that have strong genetic predisposition for T2D, including Mexican Americans, Native Americans in the Southwest, and African Americans. There is also a higher incidence of T2D in African American and Native American women than in men of the same ethnicity, suggesting a further role of sex hormones in the insulin resistance. Whether or not the person with T2D is obese or lean, prior to an increase in basal blood glucose levels, there is a loss of sensitivity to insulin so that the β-cells secrete more insulin to maintain euglycemia. Thus, in response to an oral glucose load, the prediabetic person will display up to twice the insulin secretion, and have secretion occur over a prolonged period, compared with a nondiabetic person. As the diabetes becomes more pronounced (basal hyperglycemia), insulin secretion can decrease, further compounding the hyperglycemia. In cases where T2D is precipitated by obesity, reduction in weight and increased exercise can control the hyperglycemia.

Symptoms of uncontrolled hyperglycemia include the following:

- polyuria, from the osmotic effect of glucose in the urine
- polydipsia, to compensate for urinary fluid losses
- polyphagia, because glucose cannot enter the tissues, so the tissues are "starved" and hunger is stimulated
- metabolic ketoacidosis (primarily in T1D)

Long-term uncontrolled or poorly controlled diabetes can lead to or be associated with:

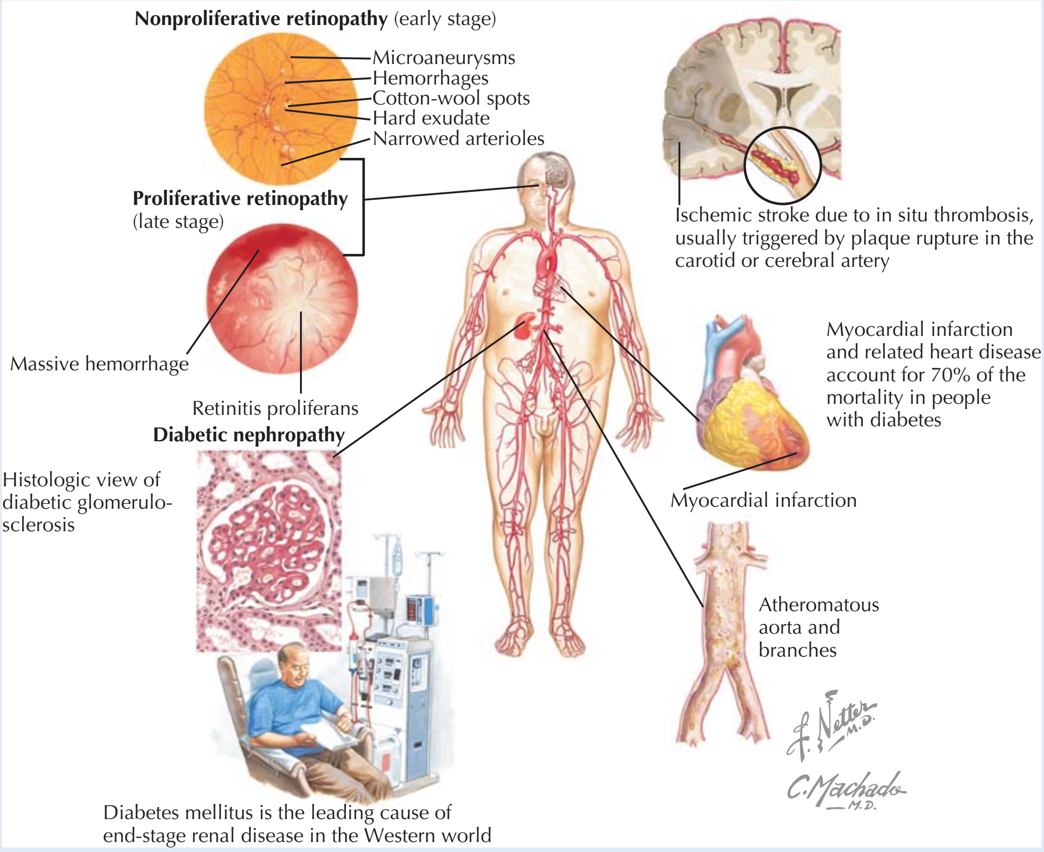
- retinopathy and blindness
- nephropathy and renal failure
- hypertension
- cerebrovascular disease
- peripheral vascular pathology

(Cont'd)

CLINICAL CORRELATE—cont'd

Whereas treatment of T1D is primarily with insulin injection (along with diet and exercise), treatment of T2D includes:

- diet and exercise, because exercise increases GLUT4 transporters in cell membranes
- insulin
- sulfonylureas and meglitinides (taken orally), which increase β-cell insulin secretion
- thiazolidinediones, which increase the sensitivity of muscle and fat tissue to insulin and decrease glucose production in the liver



Microvascular and Macrovascular Complications of Diabetes Vascular complications can occur with either type 1 or type 2 diabetes and include retinopathy (which can lead to blindness), cardiovascular disease, cerebrovascular disease, and diabetic nephropathy. These complications are responsible for the high morbidity in diabetes.

CHAPTER 30 Calcium-Regulating Hormones

OVERVIEW OF CALCIUM HOMEOSTASIS

As previously discussed in Sections 5 and 6, calcium and phosphate homeostasis are closely linked, primarily because of their roles in bone metabolism. Although several hormones regulate both minerals, this chapter will focus on calcium and its regulation. Plasma calcium concentrations are under tight control, with regulatory systems keeping normal levels ~2.4 mEq/L (or 9.4 mg/dL) of ionized calcium. Calcium is critical to a myriad of functions, including contraction of cardiac, skeletal, and smooth muscle; bone mineralization; transmission of nerve impulses; and blood clotting. Thus, alterations in plasma levels (higher or lower) can cause severe consequences (see Clinical Correlates).

Bone remodeling is a constant process involving deposition and resorption of minerals from the exchangeable calcium-phosphate mineral pool. In the bone, **osteoblasts** promote bone deposition. **Osteoclasts** are phagocytic cells that promote bone resorption (absorption of minerals out of bone and into the extracellular fluid). The activity of these two cell types are typically in balance, with bone deposition equaling bone resorption. The ability to continually remodel bone is important and contributes to increasing bone strength when bones and muscles are stressed or exercised and keeping up with bone development in growing children. When this remodeling activity decreases, as seen in the elderly, brittle bones can result.

Main Areas of Calcium Regulation

The intestines, kidneys, and bone are integral to calcium homeostasis:

- **In the GI tract:** In an average diet, adults ingest ~1000 mg of calcium daily. The intestines absorb one third of this, but because additional calcium is lost through salivary and gastrointestinal (GI) secretions, there is only about 100 to 200 mg of net calcium absorption into the blood. **1,25-dihydroxycholecalciferol** (the active form of **vitamin D**) and **parathyroid hormone (PTH)** facilitate intestinal calcium absorption.
- **In the bone:** The majority of the body's calcium (~99%) is stored in the skeletal bone, with much of the unmineralized calcium in the **osteoid,** which is an extracellular pool that is available for transport into bone or plasma. Thus, calcium readily moves between bone and plasma as part of homeostatic regulation. To a great extent, daily plasma calcium homeostasis is regulated through the effect of PTH to stimulate bone resorption, which adds calcium to the plasma. In contrast, vitamin D stimulates entry of calcium into the bone calcium pool, providing substrate for bone deposition.
- **In the kidney:** To maintain balance, because 100 to 200 mg of calcium is absorbed daily, 100 to 200 mg must be excreted by the kidneys. This represents about 2% of the filtered load of calcium to the kidneys, so 98% of the calcium is reabsorbed, in part because of stimulation by PTH.

Factors That Alter Plasma Calcium Concentration

Forty percent of plasma calcium is bound to calcium-binding proteins, mainly albumin. About 10% is bound to other anions (including bicarbonate, phosphate, and citrate), so only 50% of the calcium is free (ionized) and biologically active. The plasma provides the calcium reservoir for bone and tissue and is under tight control by calcium regulatory hormones; vitamin D; PTH; and to some extent, calcitonin. Hormone-independent changes in plasma calcium concentration can be caused by:

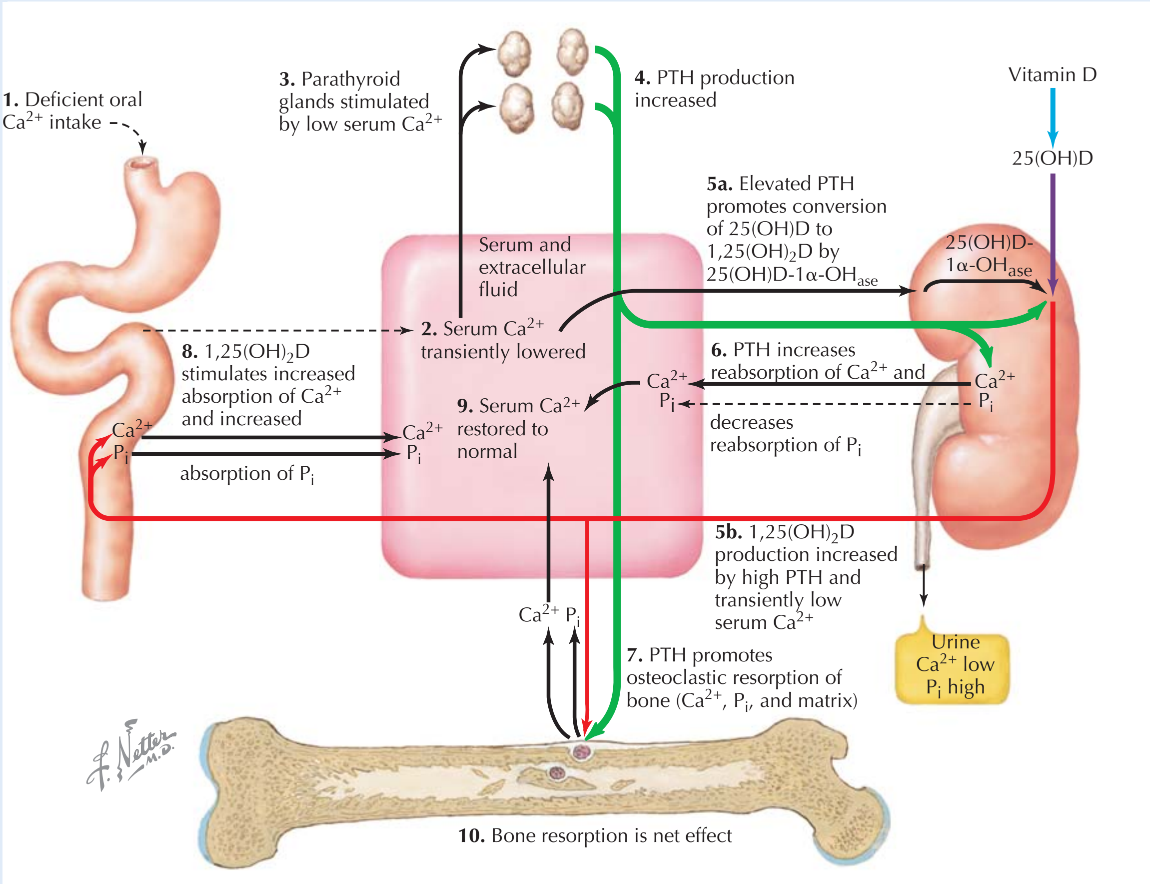
- **Acid–base disorders:** In acidosis, albumin will be used to buffer excess H^+ in exchange for Ca^{2+}, which increases free Ca^{2+} in the plasma; conversely, in alkalosis, H^+ will be released from albumin in exchange for Ca^{2+}, and plasma free Ca^{2+} will increase.
- **Changes in plasma protein concentrations:** Because 40% of plasma calcium is bound to proteins, changes in plasma proteins relate directly to total calcium: Decreases in protein will decrease total calcium, and increases in protein will increase total plasma calcium.
- **Changes in plasma anion concentrations:** This is especially relevant with plasma phosphate levels. If phosphate increases, the calcium-phosphate complexes increase, reducing ionized calcium, and vice versa. Because of this relationship, the hormone PTH regulates both calcium and phosphate.

CLINICAL CORRELATE
Effects of Altered Plasma Calcium Concentration

Hypercalcemia is an increase in plasma calcium concentration, which can have profound effects to depress nervous system activity. A 30% rise in plasma calcium (to ~12 mg/dL) will affect the nervous system, causing symptoms that can include increased urination, constipation, and important neurologic complications resulting in hyporeflexia. If plasma calcium continues to rise (above 15 to 17 mg/dL), coma and death can result.

Conversely, **hypocalcemia,** or decreased plasma calcium, will cause hyperexcitability of sensory and motor nerve and muscle cells, resulting in numbness and tingling, as well as muscle twitching and tetany. **Tetany** can occur when plasma calcium falls about 30% (to ~6 mg/dL), and death can occur if levels fall to 4 mg/dL. Hypocalcemia can result from changes in regulatory hormones, as well as dietary calcium deficiency. The effects of inadequate calcium intake are depicted in the illustration.

Changes in plasma calcium can be manifested in disorders involving parathyroid hormone and vitamin D (see "Calcium-Related Pathophysiology" Clinical Correlate).



Dietary Calcium Deficiency Inadequate dietary calcium intake *(1)* reduces plasma calcium concentration *(2),* which stimulates parathyroid hormone (PTH) secretion *(3 and 4).* PTH increases the renal production of active vitamin D *(5)* (which increases gut absorption of calcium) *(8),* increases renal calcium reabsorption and decreases renal phosphate reabsorption *(6),* and increases bone resorption of calcium and phosphate *(7).* All of these mechanisms serve to increase the plasma calcium concentration to normal *(9).* When there is a sustained calcium deficiency, the plasma calcium is maintained at a cost of severe bone demineralization *(10).*

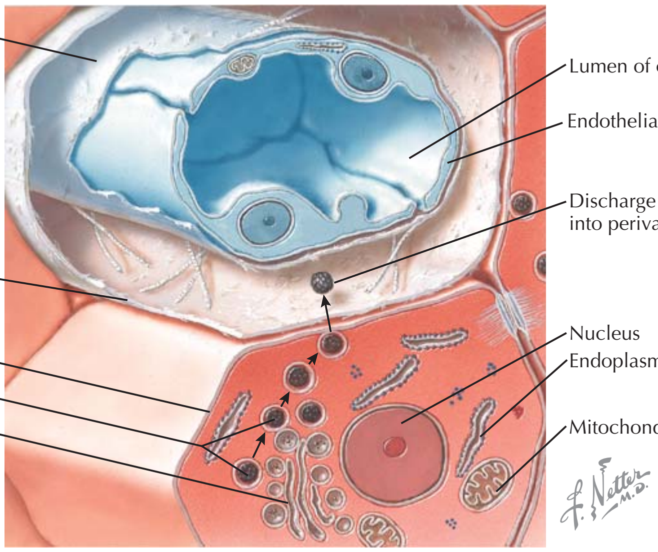


Figure 30.1 Parathyroid Hormone Secretion Parathyroid hormone is synthesized in the chief cells of the parathyroid glands. The active hormone is packaged in vesicles and stored in the cytoplasm until released. PTH secretion is stimulated by small decreases in plasma calcium levels and provides rapid mobilization of calcium into the extracellular fluid.

SYNTHESIS AND ACTIONS OF CALCIUM-REGULATING HORMONES

Parathyroid Hormone

Synthesis of Parathyroid Hormone

Parathyroid hormone (**PTH**) is the primary hormone regulating plasma calcium concentration. PTH is synthesized by the chief cells of the parathyroid glands as a 110 amino acid preprohormone (see Fig. 27.1). It is then cleaved to a prohormone in the endoplasmic reticulum and to the active 84 amino acid peptide hormone in the Golgi apparatus. It is then packaged into secretory vesicles in the cytoplasm, and the vesicles stored until needed (Fig. 30.1).

Actions of Parathyroid Hormone

The hormone is continually synthesized, and there is constant low-level release of PTH into the blood. In response to small decreases in ionized calcium in the plasma, additional PTH is released into the blood and acts at the intestines, bone, and kidney to restore plasma calcium to normal levels (Fig. 30.2). PTH stimulates cAMP through its G protein–coupled receptors:

- **At the kidney:** (1) to increase calcium reabsorption at the *distal tubule,* and to inhibit phosphate reabsorption at both the proximal and distal tubules; and (2) to increase the synthesis of 1,25-dihydroxycholecalciferol, the active metabolite of vitamin D. This conversion is critical to producing active vitamin D, and loss of PTH or loss of kidney function can impair vitamin D metabolism and function. The actions at the kidney are rapid and are a main avenue for the rapid restoration of plasma calcium levels.
- **At the bone:** To increase bone resorption, freeing calcium into the blood. This occurs by **osteocytic osteolysis,** which rapidly (within 1 to 2 hours) mobilizes calcium from the unmineralized calcium phosphate pool, and by **osteoclastic osteolysis,** which breaks down mineralized bone over about 12 hours. In response to more severe, prolonged decreases in plasma calcium (days to weeks), PTH can stimulate the proliferation of osteoclasts, to further break down mineralized bone. The end result is increased calcium and phosphate in the plasma, at the expense of loss of mineralization of bone.

The chief cells of the parathyroid gland have calcium sensors, which monitor plasma calcium concentration. This allows rapid response when plasma calcium decreases, and PTH can quickly mobilize calcium from bone and stimulate vitamin D production. Thus, PTH is crucial to the maintenance of calcium homeostasis.

Parathyroid hormone–related protein (**PTHrp**) is a protein that mimics the calcium-mobilizing effects of PTH. It is present in a variety of tissues including smooth muscle, breast, and the central nervous system and is believed to act in a paracrine manner. Although its normal actions are still being investigated, PTHrp is also secreted from certain tumors, such as breast cancers and renal cell and squamous cell carcinomas.

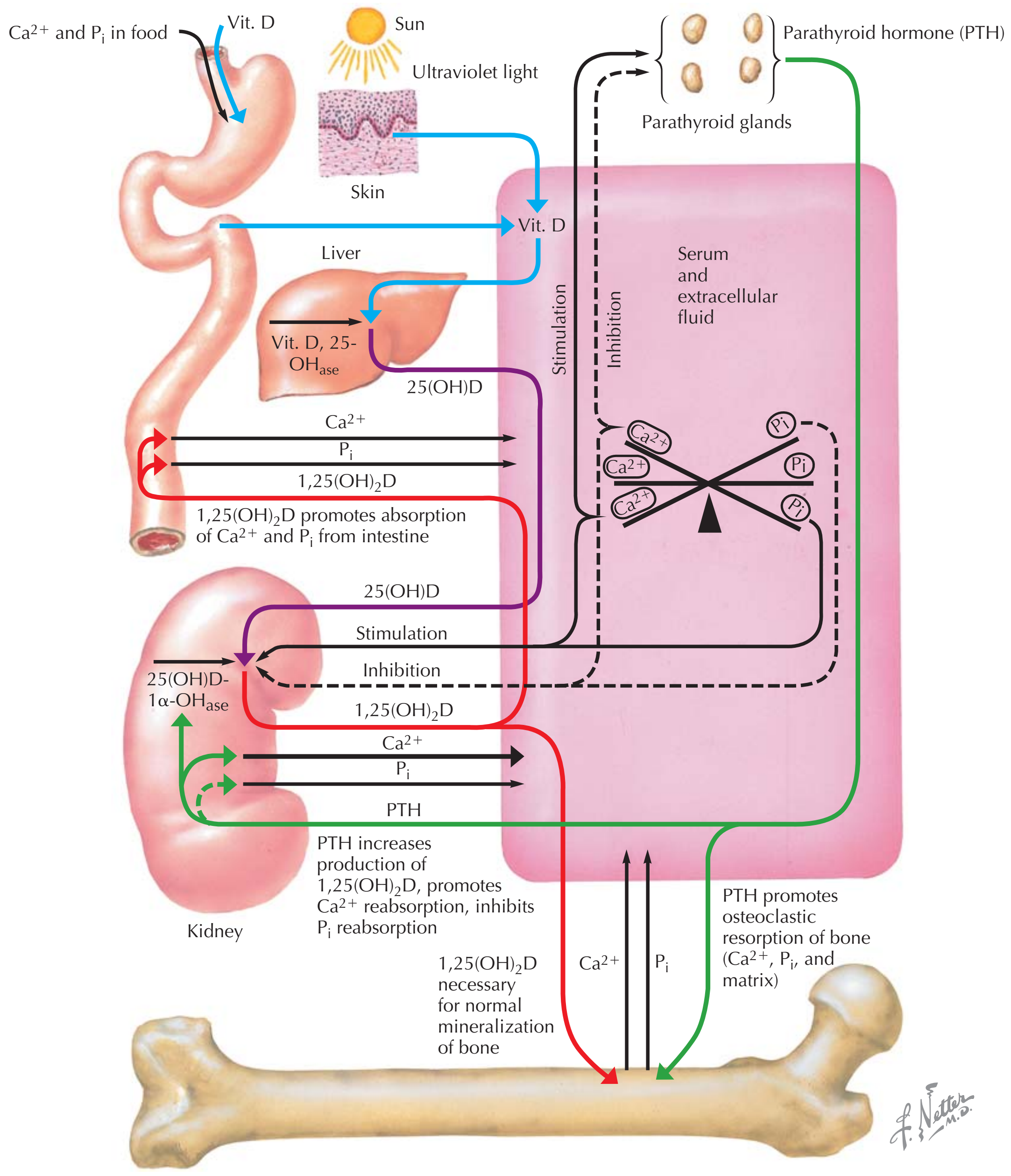


Figure 30.2 Parathyroid Hormone and Vitamin D Actions on Plasma Calcium Concentrations Parathyroid hormone (PTH) is secreted from the parathyroid glands in response to a decrease in plasma ionized Ca^{2+}. PTH acts rapidly (1) on bone to cause resorption and increase plasma Ca^{2+}; and (2) on the kidney to increase Ca^{2+} reabsorption and decrease phosphate reabsorption *and* increase production of active vitamin D. The increase in vitamin D increases Ca^{2+} absorption in the gut and promotes bone mineralization. Overall, the rapid effects of PTH increase Ca^{2+} in the blood, restoring homeostasis.

Vitamin D

Synthesis of Vitamin D

Vitamin D, or **cholecalciferol,** is a sterol that enters the blood and extracellular fluid (ECF) from dietary sources and from the action of ultraviolet light on 7-dehydroxycholesterol in skin cells. The initial form of vitamin D is converted to 25-hydroxycholecalciferol in the liver and then undergoes final hydroxylation to the active form, **1,25- dihydroxycholecalciferol,** in the kidneys. Active vitamin D is bound to plasma proteins and acts to increase plasma calcium concentration and promote bone mineralization.

Actions of Vitamin D

When plasma calcium levels decrease, PTH increases and stimulates renal 1-α-hydroxylase to increase formation of active vitamin D. The PTH will also increase resorption of calcium-phosphate from bone and increase renal calcium reabsorption, with the total effect of increasing plasma calcium levels. Active vitamin D binds to its intracellular steroid receptor and acts:

- **At the small intestine:** To increase calcium absorption by increasing apical **calbindin** (Ca^{2+}-binding proteins) and increasing the basolateral Ca^{2+} ATPases. This enhances intestinal absorption of calcium. Vitamin D also increases phosphate absorption, and excess plasma phosphate is excreted by the kidneys (by PTH action).
- **At the bone:** To increase bone remodeling and mineralization through effects on both osteoblasts and osteoclasts. At high concentrations, vitamin D can cause bone demineralization.

The synthesis and actions of vitamin D take time, but they act to increase the supply of calcium from the diet. Thus, whereas in the short-term plasma calcium levels are regulated by PTH, the activation of vitamin D by PTH is important in long-term calcium homeostasis. When calcium levels are stabilized, calcium can be restored to the bone by the vitamin D.

Active vitamin D levels in the blood are tightly regulated by feedback control. Under normal conditions, 25-hydroxycholecalciferol concentrations in the plasma will remain constant, despite large increases in dietary vitamin D intake. In addition, the 25-hydroxycholecalciferol form can be stored in the liver for several months and released and converted to active vitamin D when needed, allowing another layer of regulation for plasma levels of vitamin D. This overall control of vitamin D levels is important, because overproduction of active vitamin D can actually cause bone resorption, instead of deposition.

Calcitonin

Calcitonin is a 32 amino acid peptide hormone that is produced in the parafollicular cells of the thyroid gland, and although it appears to have effects on calcium handling, it does not contribute to the rapid regulation of calcium. It is released in response to elevated plasma calcium levels and acts on G protein–coupled receptors:

- **At the kidney:** To increase both calcium and phosphate excretion.
- **At the bone:** To decrease bone resorption by reducing osteoclastic activity.

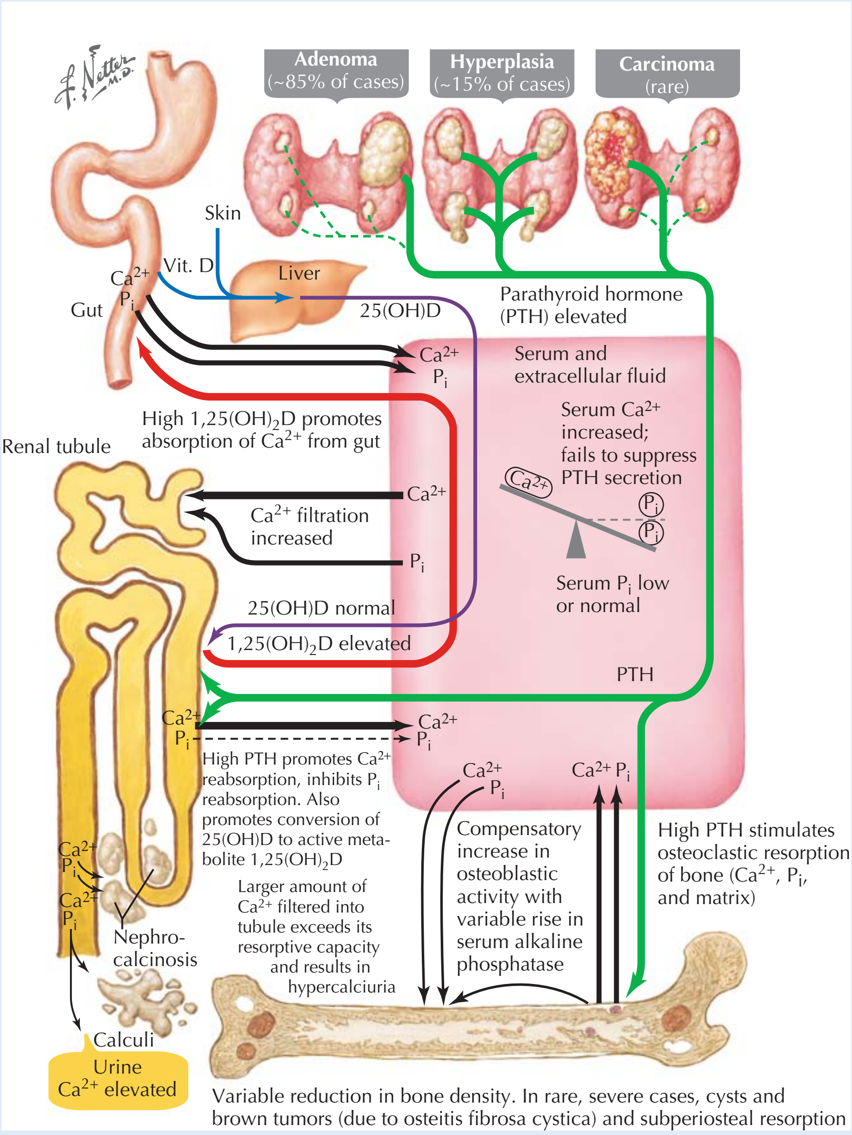
It is not clear whether the physiologic effects of calcitonin are important in humans, because removal of the parafollicular cells has no dramatic effect on calcium homeostasis.

CLINICAL CORRELATE
Calcium-Related Pathophysiology: Hyperparathyroidism

Hyperparathyroidism is associated with excessive PTH secretion and results in hypercalcemia (high plasma calcium) and hypophosphatemia (low plasma phosphate).

Primary hyperparathyroidism is usually caused by parathyroid tumors that produce PTH but do not have feedback regulation. The elevated PTH increases the renal activation of vitamin D; increases plasma calcium through increasing intestinal absorption, kidney reabsorption, and bone resorption; and decreases plasma phosphate by reducing renal phosphate reabsorption. Although renal calcium reabsorption is elevated, the significantly increased filtered calcium load results in elevated urinary calcium and phosphate excretion, increasing the chance of kidney stone formation. Though many patients are asymptomatic, the elevation in plasma calcium can cause some or all of the following "stones," "bones," "groans," and "moans": kidney stones from calcium-phosphate and calcium-oxalate formation in urine, bone pain from excessive resorption, groans from constipation due to reduced fecal calcium, and moans from generally not feeling well and fatigue. Treatment is surgical removal of the tumor.

Secondary hyperparathyroidism is caused in response to the primary problem of low plasma calcium levels. In secondary hyperparathyroidism, the low plasma calcium levels are usually caused by vitamin D deficiency and/or chronic renal failure, and the parathyroid glands secrete PTH to correct this problem. Thus, PTH levels will be high, but plasma calcium will be low or normal. Treatment is with vitamin D, when indicated.



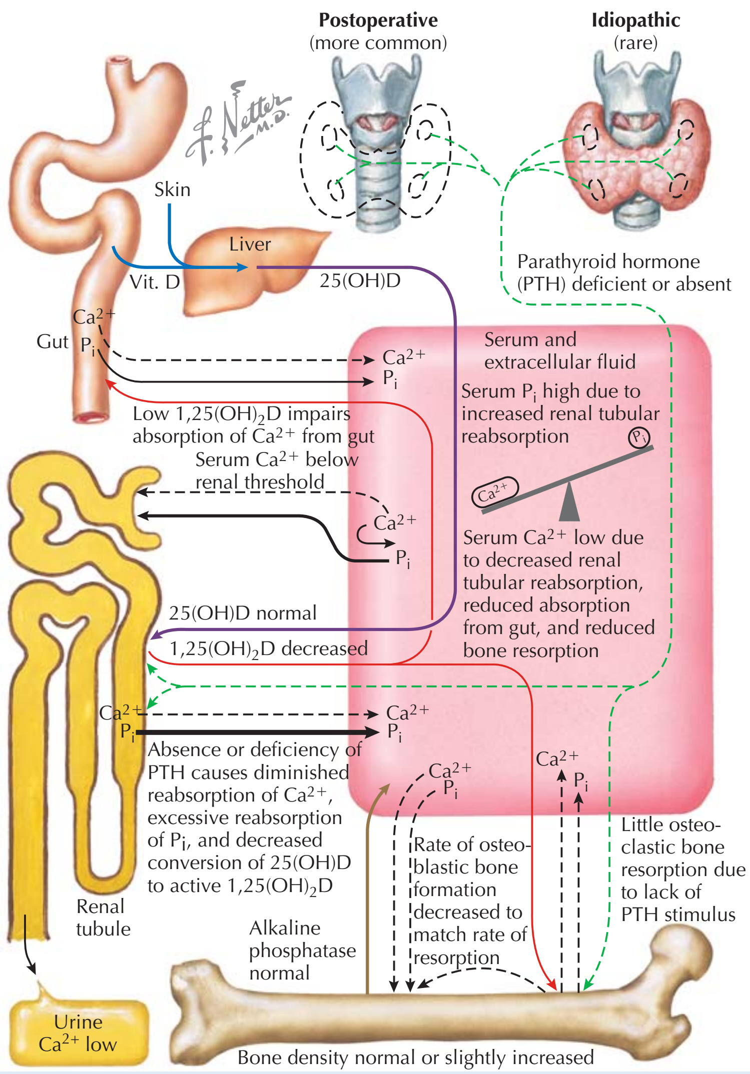
Pathophysiology of Hyperparathyroidism Most cases of hyperparathyroidism arise from parathyroid tumors that cause hypersecretion of parathyroid hormone (PTH). The elevated PTH increases bone resorption, renal calcium reabsorption, and active vitamin D production (which increases intestinal calcium absorption). This causes hypercalcemia, which can result in multiple symptoms ("stones," "bones," "groans," and "moans").

CLINICAL CORRELATE

Calcium-Related Pathophysiology: Hypoparathyroidism

Hypoparathyroidism is a deficiency of PTH, and results in hypocalcemia (low plasma calcium) and hyperphosphatemia (high plasma phosphate). This is a fairly common occurrence following thyroid or parathyroid surgery, and the decreased extracellular fluid calcium increases the excitability of sensory and motor neurons and muscle cells. This excitability can cause sensory effects such as tingling or numbness in the lips and fingers, as well as motor effects including cramping and twitching. Severe hypocalcemia can lead to tetanic muscle spasms and seizures, so treatment is initiated when first symptoms are noted. Initial treatment can include intravenous calcium (if plasma levels are dangerously low), followed by oral calcium supplements with active vitamin D. Untreated, chronic hypoparathyroidism can result in cataracts, alopecia (hair loss), and weakened tooth enamel, in addition to the muscle spasms and seizures. Although rare, sustained low (4 mg/dL) plasma calcium concentrations can lead to coma and death. If life-threatening, treatment is with intravenous calcium; otherwise, active vitamin D and oral calcium is effective in maintaining plasma calcium levels.

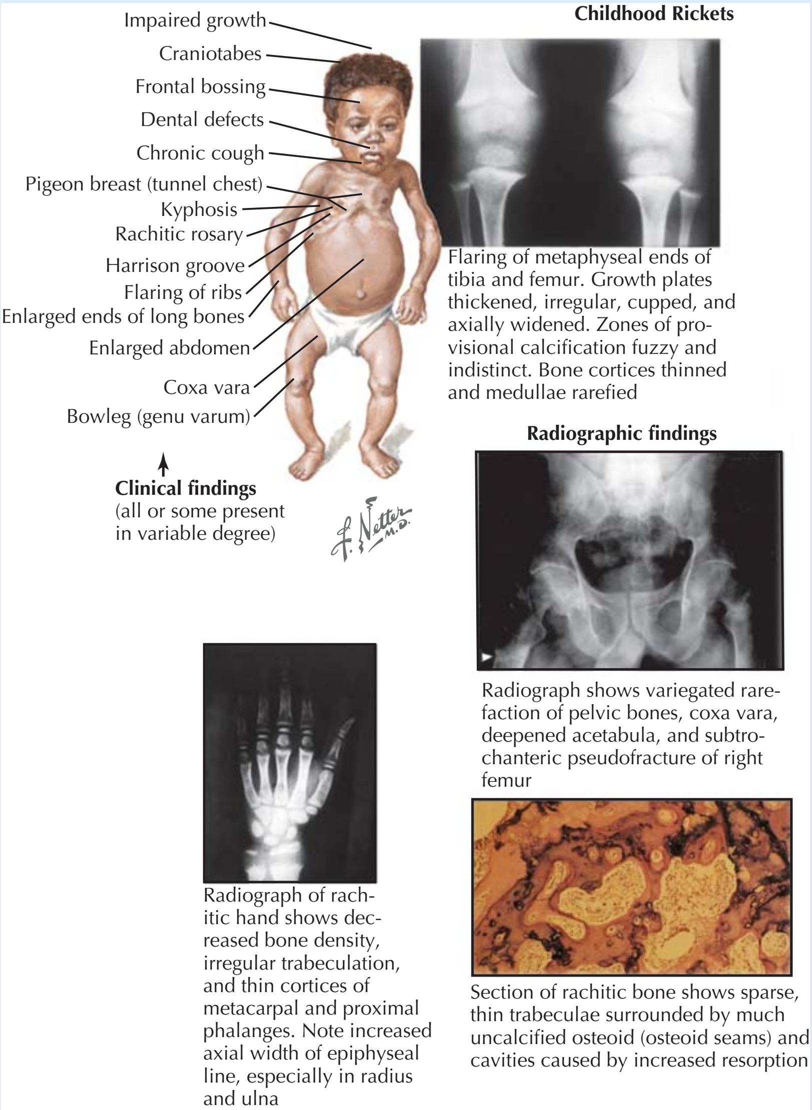
Pseudohypoparathyroidism has the same overall effect as hypoparathyroidism, reducing plasma calcium and increasing plasma phosphate—however, the cause is a primary defect in the G_s protein of the PTH receptors in bone and kidney. Although PTH can bind, the cAMP-pathway cannot be activated, and thus while PTH levels are *elevated*, renal and bone calcium transport is not activated.



Pathophysiology of Hypoparathyroidism Hypoparathyroidism is usually a result of thyroid or parathyroid surgery, and the lack of parathyroid hormone reduces the formation of active vitamin D, decreases renal calcium reabsorption, and alters bone resorption activity. Hypocalcemia develops, and the low plasma calcium levels increase neuronal and muscle cell excitation, causing twitching, cramping, and in extreme cases tetany.

CLINICAL CORRELATE
Calcium-Related Pathophysiology: Vitamin D Deficiency

Vitamin D deficiency is rarely found in areas where vitamin D is supplemented in the diet and there is regular exposure to sunlight. **Rickets** is a disease caused by insufficient vitamin D during childhood. The lack of vitamin D prevents adequate calcium and phosphate for bone mineralization, resulting in stunted growth and malformed bones. In adults, prolonged vitamin D deficiency will decrease new bone mineralization, resulting in bone deformation **(osteomalacia)**. This is especially evident in weight-bearing bones, and can cause severe bow-legged stance. Although it is rare to see this in adults, it can result from chronic steatorrhea, because vitamin D is fat-soluble, and most will be lost in the feces. Renal rickets is a specific form of osteomalacia caused by chronic renal damage. This limits the production of vitamin D, because the kidney tissue is either damaged or absent, and PTH cannot convert 25-hydroxycholecalciferol to active 1,25-dihydroxycholecalciferol.



Rickets The main manifestation of vitamin D deficiency is in bone mineralization. As depicted in the illustration, in children, this results in rickets. In adults, vitamin D deficiency leads to osteomalacia.

CHAPTER 31

Hormones of the Reproductive System

Sex hormones are involved in fetal development and sexual differentiation, pubertal somatic changes, and reproductive physiology. Although the term **sex hormone** refers to sex steroids (primarily estrogen, progesterone, testosterone, and adrenal androgens), a wider array of hypothalamic, pituitary, gonadal, and placental hormones have important roles in the reproductive system. Endocrine problems involving the reproductive system are relatively common. For example, infertility, often a result of an endocrine abnormality, affects approximately 10% of the population.

Synthesis and secretion of testosterone by the fetal testes is stimulated initially by a placental hormone known as **chorionic gonadotropin,** the same hormone that is essential in maintaining maternal ovarian progesterone production during pregnancy. After the first trimester, fetal pituitary **luteinizing hormone** (**LH**) is required to maintain androgen production in a male fetus. Similarly, in a female fetus, estrogens are secreted by the ovaries in response to pituitary LH. After birth, the pituitary gonadotrophs become quiescent until puberty. The role of gonadotropins in pubertal development, reproductive function, and pregnancy is discussed below.

FETAL DEVELOPMENT OF THE REPRODUCTIVE ORGANS AND DIFFERENTIATION OF GENITALIA

An individual's sex can be characterized in terms of genetics (the male XY genotype vs. the female XX genotype), gonads (testes vs. ovaries), and phenotype (external appearance of "maleness" vs. "femaleness"). Although genetic sex is determined at the time of conception, the fetus remains undifferentiated in terms of gonadal sex for the first 5 weeks of development (Fig. 31.1).

Development of Gonadal Sex

Under the control of the **SRY gene** on the Y chromosome, the undifferentiated gonads develop into **testes;** full development of gonadal sex requires several other genes on the Y chromosome and various autosomal chromosomes. By 8 to 9 weeks of development, the **Leydig cells** of the testes begin to secrete **testosterone.** Development of **ovaries,** on the other hand, begins at week 9 of gestation and requires the presence of two X chromosomes. Germ cells within the ovaries give rise to **oogonia,** which proliferate and soon enter meiosis, leading to the **primary oocyte stage,** at which meiosis is arrested until activated during sexual cycles after puberty. The developing ovary produces the **estrogens.**

During the undifferentiated stage of fetal development, two pairs of ducts, the **wolffian ducts** and the **müllerian ducts,** develop in both sexes (see Fig. 31.1). Under the influence of testosterone, the wolffian ducts develop into male structures including the epididymis, vas deferens, and seminal vesicles. Meanwhile, **müllerian inhibitory factor,** a dimeric glycoprotein hormone secreted by the **Sertoli cells** of the fetal testes, causes regression of the müllerian ducts. The combination of these two series of events leads to the development of the male system. In the absence of testes and their secretions, the müllerian ducts persist and develop, forming the fallopian tubes, uterus, and the upper vagina, whereas the wolffian ducts degenerate.

Development of Genital Sex

The external genitalia are also present in an undifferentiated form during early development (Fig. 31.2). Differentiation begins at 9 to 10 weeks of gestation. In the absence of androgens, female genitalia are produced. In the male fetus, testosterone produced by the testes and secreted into the circulation is converted to **dihydrotestosterone** (**DHT**) in the primitive genital structures; DHT stimulates the formation of the male genitalia (conversion of testosterone to DHT is not involved in the actions of testosterone on wolffian duct differentiation).

Phenotypic Sex

Phenotypic sex is the outward appearance of maleness or femaleness. Whereas genetic sex is based on the presence of the XY or XX genotype, and gonadal sex is determined by specific genes on the X chromosome as well as several autosomal genes, phenotypic sex develops in response to gonadal hormones. In the presence of testosterone, male genitalia develop; in its absence, the female form of genitalia prevail. Other phenotypic characteristics of femaleness are promoted

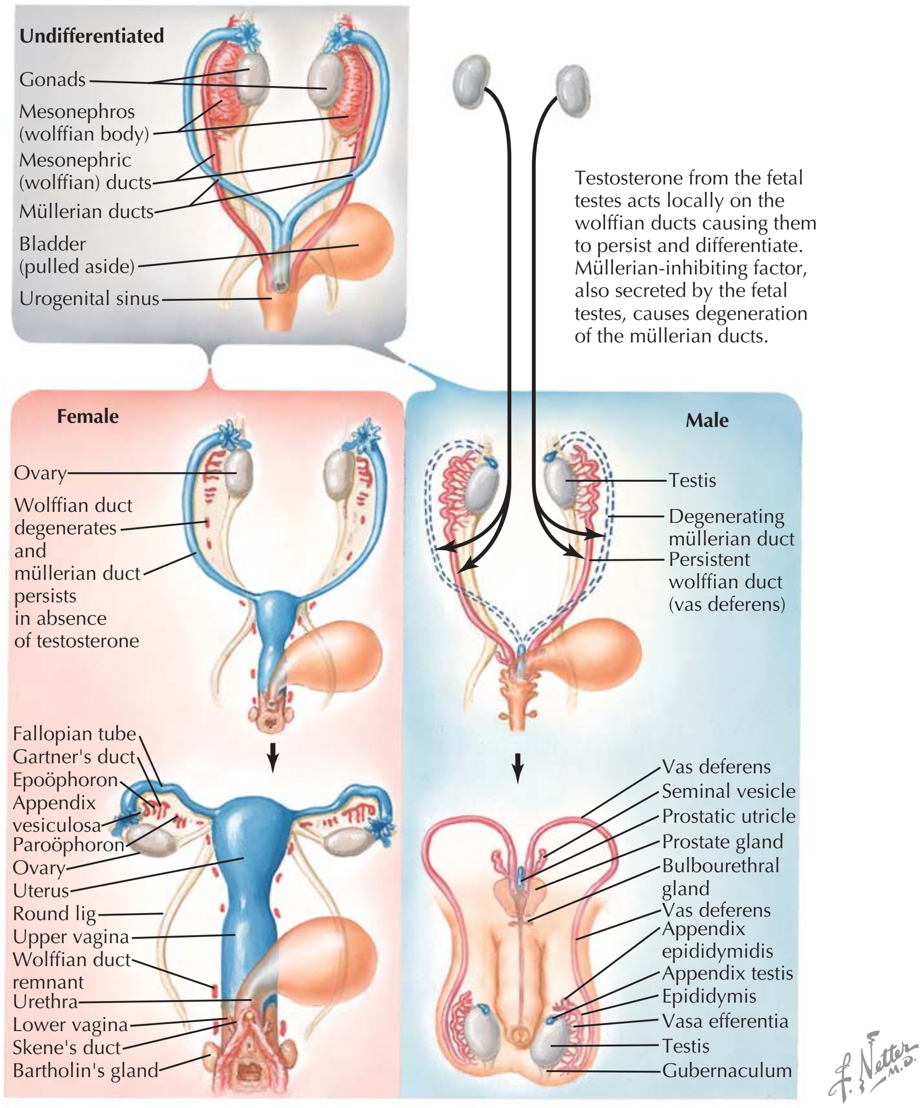


Figure 31.1 **Gonad and Genital Duct Formation** The undifferentiated gonads and ducts of the early embryo differentiate into the male or female gonads and duct systems under the influence of various products encoded by the X and Y chromosomes. Notably, a product of the SRY gene of the Y chromosome results in differentiation of the gonads into testes. Production of testosterone by the testes results in persistence and differentiation of the wolffian ducts. Müllerian-inhibiting factor secreted by the testes causes müllerian duct degeneration. In the female fetus, in the absence of testosterone the gonads develop into ovaries and the wolffian ducts degenerate.

by the absence of androgens and presence of estrogen. In androgen insensitivity syndrome, for example, androgen receptors are absent, and the female phenotype prevails.

PUBERTY

Beginning a year or two before the onset of puberty, production of androgens by the zona reticularis of the adrenal glands commences. The major adrenal androgens are **dehydroepiandrosterone** (DHEA) and **androstenedione.** This event is known as **adrenarche** and is distinct from puberty. Secretion of adrenal androgens results in:

- appearance of pubic and axillary hair
- adult body odor
- acne and oiliness of skin

Hormonal Regulation of Puberty

The actual process of **puberty** is associated with maturation of the hypothalamus and anterior pituitary gland. The hypo-

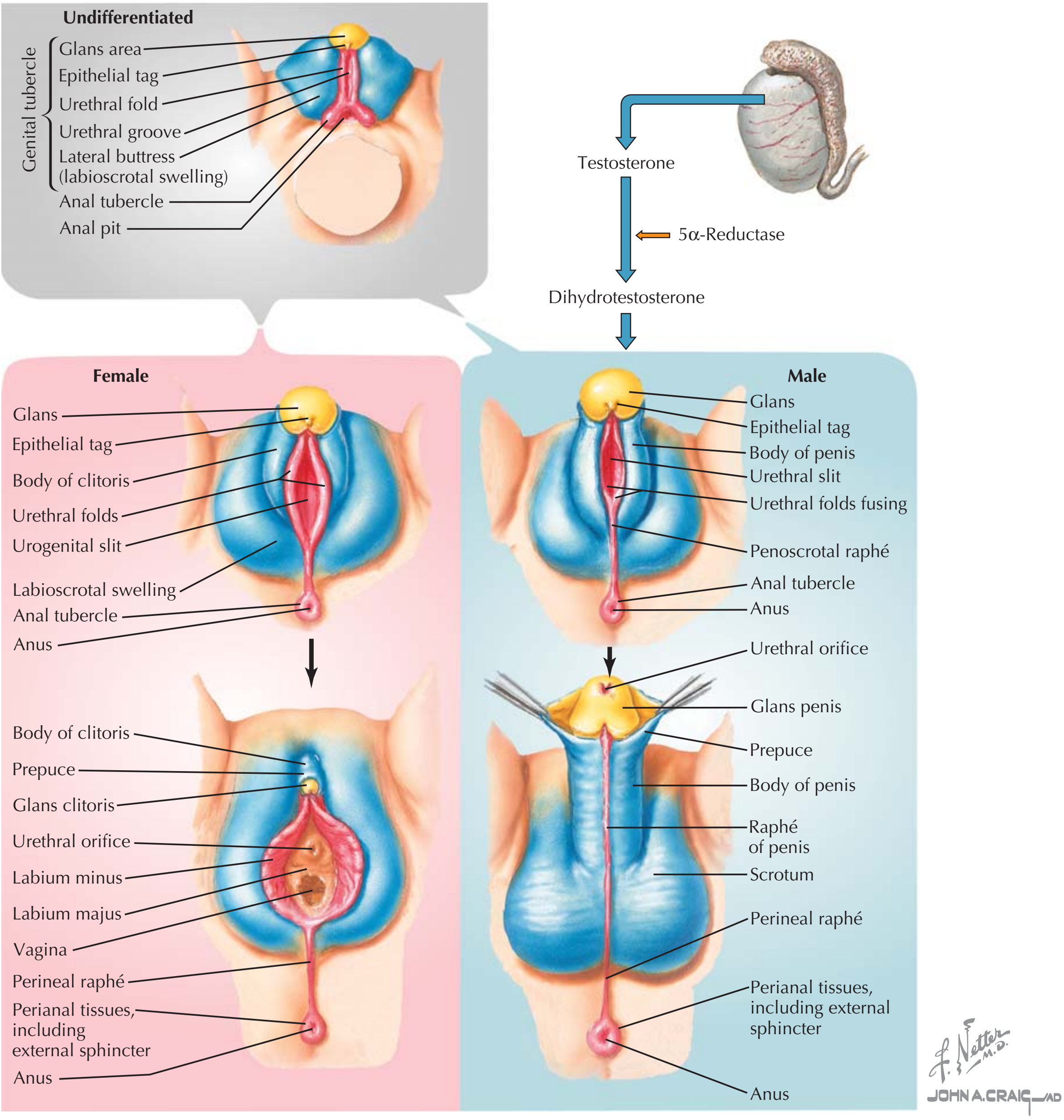


Figure 31.2 **Differentiation of the External Genitalia** In the absence of testosterone, the undifferentiated external genitalia develop into the female structures. Testosterone, after conversion to dihydrotestosterone, stimulates the formation of male external genitalia from the undifferentiated structures. Homologies between male and female genital structures are color-coded.

thalamic secretion of the decapeptide **gonadotropin-releasing hormone** (GnRH) increases and becomes pulsatile, as the hypothalamus becomes less sensitive to negative feedback by sex steroids. Thus, a pulsatile pituitary secretion of the gonadotropins **follicle-stimulating hormone** (FSH) and **luteinizing hormone** (LH) is established, stimulating gonadal maturation and sex hormone synthesis. **Estradiol** (the major estrogen or "estrogenic hormone") and **progesterone** are the predominant sex hormones produced in women, whereas **testosterone** is the main sex steroid of men (the details of the hypothalamic-pituitary-gonadal axis are discussed later in the chapter). Adrenal androgens continue to have a physiological role in puberty and beyond in women (effects of adrenal androgens are overshadowed by those of testosterone in men). In males, some testosterone is converted to estradiol, and the latter hormone has important functions in male puberty. The

The pubertal growth spurt typically begins earlier in girls than in boys. Previously, it was believed that androgens (testosterone and adrenal androgens) were responsible for the rapid increase in height. Recent studies, however, have demonstrated that although testosterone increases bone mass and density, estradiol is the hormone primarily responsible for the growth spurt in both sexes. Estradiol stimulates long bone growth but also promotes closure of the growth plates. Females, who enter puberty earlier, attain less adult height than males.

physiological and anatomical changes occurring at puberty are the result of various hormone actions and include:

- Gonadal maturation and sex steroid synthesis and release, as a result of increased gonadotropin secretion.
- Maturation of other sex organs and genitalia, mainly stimulated by estradiol in females and testosterone in males.
- Development of secondary sexual characteristics (characteristics of "maleness" or "femaleness" not directly associated with reproduction). These include gender-specific changes in distribution of body and facial hair, breast development, body fat distribution, muscle development, and pitch of voice. Estrogen is primarily responsible for female characteristics, whereas testosterone is the major stimulus for development of male secondary sexual characteristics.
- The linear growth spurt. This event is attributed mainly to estradiol in both sexes; estrogenic hormones cause accelerated growth of long bones but also promote the eventual closure of the epiphyseal plates of these bones, and thus the end of growth in height.

THE MENSTRUAL CYCLE AND FEMALE REPRODUCTIVE ENDOCRINOLOGY

In late puberty, adolescent girls experience their first menstruation, an event known as **menarche,** and **menstrual cycles** commence, continuing until **menopause** (unless interrupted by pregnancy). These cycles last an average of 28 days and consist of three phases (Fig. 31.3):

- The **follicular phase,** characterized by proliferation of the endometrium of the uterus and development of ovarian follicles.
- The **ovulatory phase,** during which one follicle that has fully matured ruptures and releases an ovum.
- The **luteal phase,** characterized by transformation of follicular cells into a **corpus luteum** and further proliferation of the endometrium. Unless implantation of a fertilized ovum takes place, the corpus luteum regresses and menses follows, during which the proliferated endothelium is sloughed off and bleeding occurs for a period of 3 to 5 days.

Follicular Phase

The development of follicles within the ovary is illustrated in Figure 31.4. During each cycle, after the onset of menses (by convention, "day 1" of the cycle), several primordial ovarian follicles begin to undergo further development under the influence of FSH, and hence, the term **follicular phase** is used to describe the first half of the cycle. Within the developing follicles, **theca interna** cells secrete **androgens,** which are converted to **estradiol** by the **granulosa cells** of the follicles. This conversion is stimulated by LH. Estradiol causes endometrial proliferation, as well as development of glands and growth of spiral arteries within the endometrium, in preparation for possible implantation of a fertilized egg. For this reason, the follicular phase is also called the **proliferative phase.** In addition, estradiol promotes secretion of watery cervical mucus, through which sperm can enter the uterus. Ultimately, one of the developing follicles predominates and becomes a **mature follicle** (graafian follicle), and the others regress.

Ovulatory Phase

Estradiol exerts negative feedback on the hypothalamic secretion of GnRH and anterior pituitary secretion of FSH (LH is not suppressed during this period) through much of the follicular phase. Additionally, granulosa cells of developing follicles secrete a peptide hormone, **inhibin,** which has negative feedback effects specifically on FSH. Toward the end of the follicular phase, estradiol rises to a level at which **positive feedback** is triggered (Fig. 31.5). A surge in LH, and to a lesser extent, FSH, takes place and produces **ovulation** at midcycle, releasing a mature ovum, which is carried by ciliary action into the fallopian tube (see Fig. 31.3). Interestingly, a mature ovum is produced in alternating ovaries from month-to-month, but if a woman has only one functional ovary, that one ovary will normally produce a mature ovum monthly.

Luteal Phase

In the ensuing luteal phase of the cycle, the ruptured follicle undergoes involution, forming the **corpus luteum.** Progesterone and inhibin production by the corpus luteum rise, as does estradiol production to a lesser degree. Estrogens, progesterone, and inhibin now contribute to negative feedback on the hypothalamus and anterior pituitary (see Fig. 31.5). Further proliferative and secretory changes take place in the endometrium, stimulated by progesterone; the luteal phase is also called the **secretory phase** for this reason. In the cervix, secretions become thicker, making passage of sperm into the uterus more difficult. Conception must take place within a day or two of ovulation, because the ovum is viable for only a short period after release from the graafian follicle (normally, conception takes place while the egg is in transport within a fallopian tube). Toward the end of the luteal phase, unless pregnancy occurs, steroid and inhibin secretion fall, and menses results.

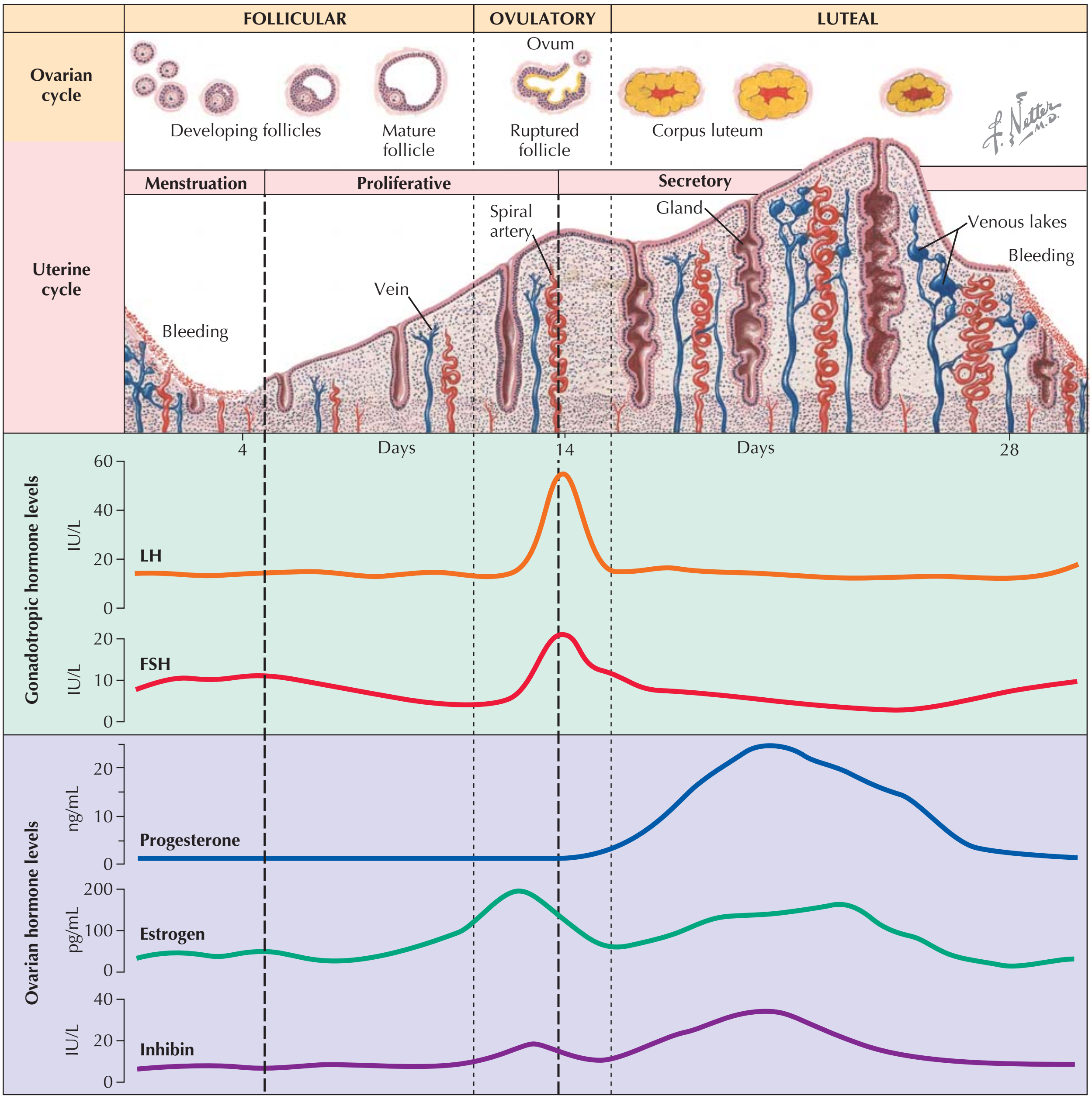


Figure 31.3 **Menstrual Cycle** During the female menstrual cycle, changes take place in the ovaries and uterus, under the control of the hypothalamus and anterior pituitary gland. During the follicular phase, several primary follicles undergo further development in response to FSH and synthesize androgens, which are converted to estradiol under the influence of LH. Ultimately, one follicle fully matures and the others regress. The uterine endometrium proliferates in response to estradiol. Near midcyle, estradiol rises to a level that initiates *positive feedback*, and thus a surge in LH and FSH release by the anterior pituitary, which results in ovulation. During the ensuing luteal phase, the mature follicle becomes the corpus luteum, which secretes progesterone and estradiol. The uterus undergoes further proliferative and secretory changes. Unless pregnancy occurs, endometrial sloughing and menstruation eventually occur, marking the beginning of a new cycle. FSH, follicle-stimulating hormone; LH, luteinizing hormone.

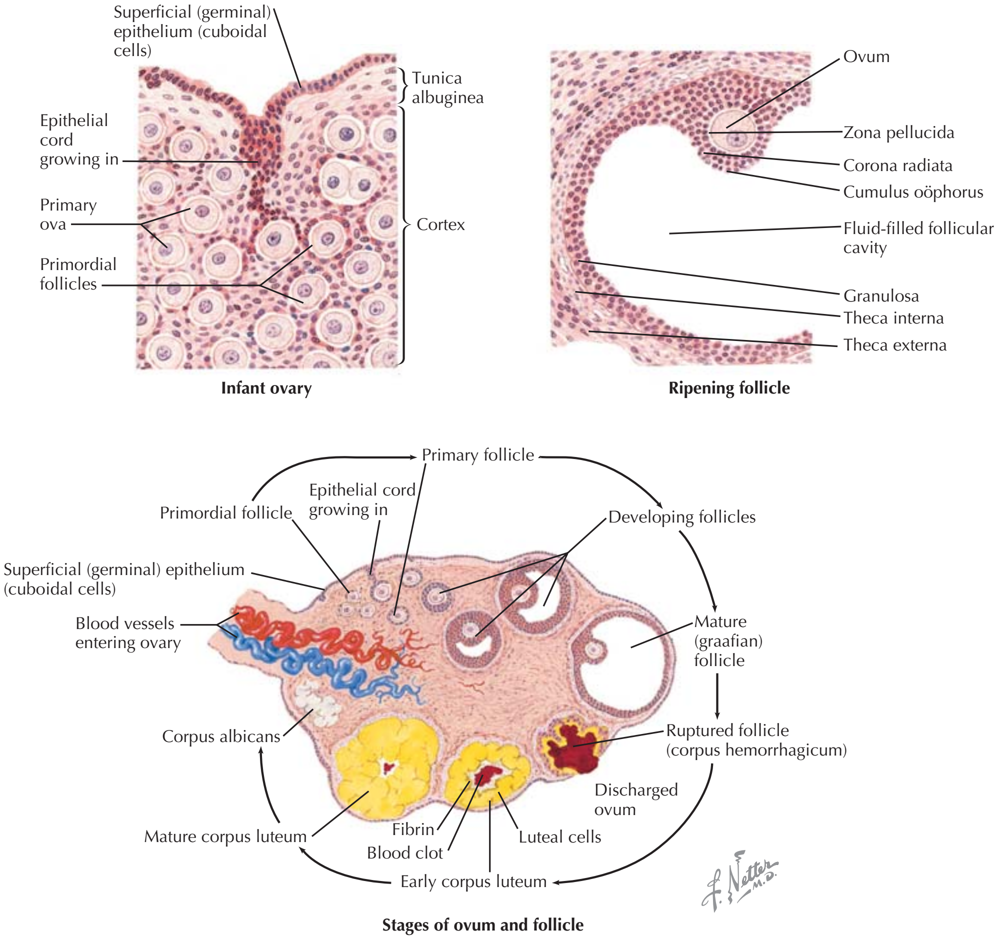


Figure 31.4 **Ovary, Ova, and Follicles** Until puberty, the ovary contains numerous primordial follicles that remain in a dormant state. After puberty, several follicles begin ripening with each menstrual cycle, in stages illustrated in the bottom panel. Only one follicle becomes a mature follicle; the others ultimately regress. After ovulation and release of the ovum, the mature follicle involutes to form the corpus luteum, which persists to the end of the cycle.

CLINICAL CORRELATE
The Rhythm Method and Oral Contraceptives

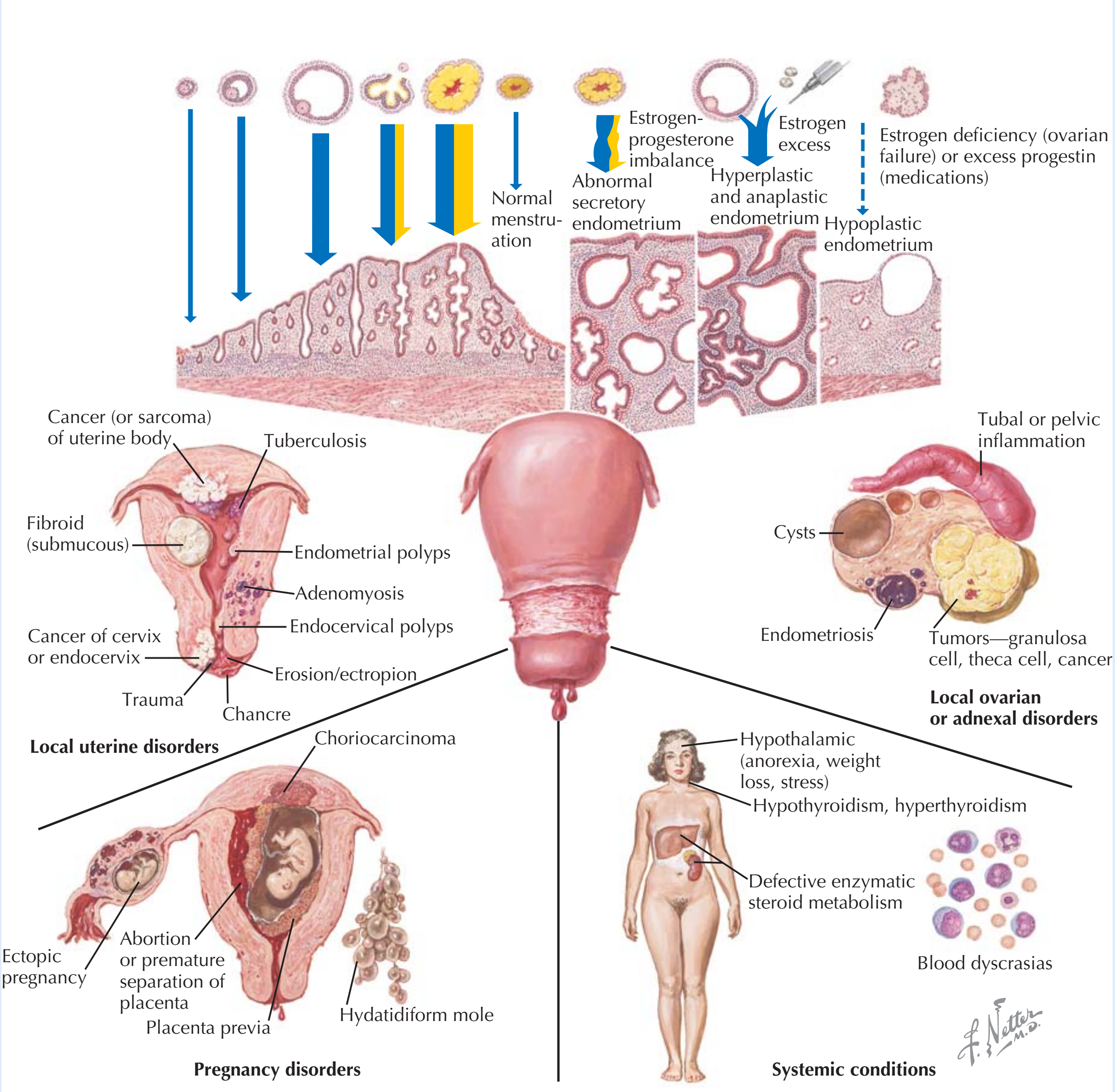
The average menstrual cycle is 28 days in length (beginning with the first day of menses), but this may vary both between women and for an individual woman. Ovulation takes place at about midcycle, but more precisely, it occurs *14 days before the onset of the next menses*. In other words, in a 26-day cycle, ovulation takes place on day 12, but in a 30-day cycle, ovulation takes place on day 16. Sperm are viable in the female reproductive tract for a few days, while mature ova are viable for only a short time after release.

For pregnancy to occur, intercourse needs to take place between approximately 5 days before ovulation and at most a day after ovulation. However, because it is not possible to predict exactly when ovulation will take place (because the date of onset of the next menses cannot be predicted with certainty), "rhythm methods" of contraception have low reliability compared with most forms of birth control. In various rhythm methods, intercourse is avoided for several days before and after the predicted date of ovulation. For example, in the Standard Days Method, women whose cycles are between 26 and 32 days in length avoid intercourse between days 8 and 19 of their cycles. In general, the rhythm method has a failure rate of several percent annually when used perfectly, and up to 25% otherwise.

With the use of oral contraceptives, most commonly a combination of an estrogen and a progestin (progesterone-like drug), cycles are controlled by the drug. Typically, hormone-containing pills are taken for 21 days, followed by a week of no pills or daily sugar pills. Pregnancy is prevented primarily by inhibition of ovulation, due to inhibition of gonadotropin release by the oral hormones. Bleeding (menstruation) occurs upon withdrawal from the hormones after 21 days. With perfect use, the annual rate of conception is 0.3%, although the failure rate is actually several percent annually.

CLINICAL CORRELATE
Abnormal Uterine Bleeding

Normal menstrual bleeding follows the luteal phase of the cycle and is caused by the fall in gonadal hormone levels. Heavy bleeding during menses (menorrhagia) or irregular bleeding during the cycle may be caused by hormonal imbalance. Abnormal uterine bleeding can also be caused by a variety of uterine or adnexal disorders as well as systemic disorders in nonpregnant, reproductive-age women. In pregnant women, bleeding may be caused by rupture of a placental blood vessel or may foreshadow an impending miscarriage. Placenta previa, in which the placenta extends over the cervix, is often associated with bleeding after the first trimester of pregnancy. Various causes of abnormal uterine bleeding are illustrated.



Causes of Abnormal Uterine Bleeding Abnormal uterine bleeding is associated with a variety of disease processes and disorders.

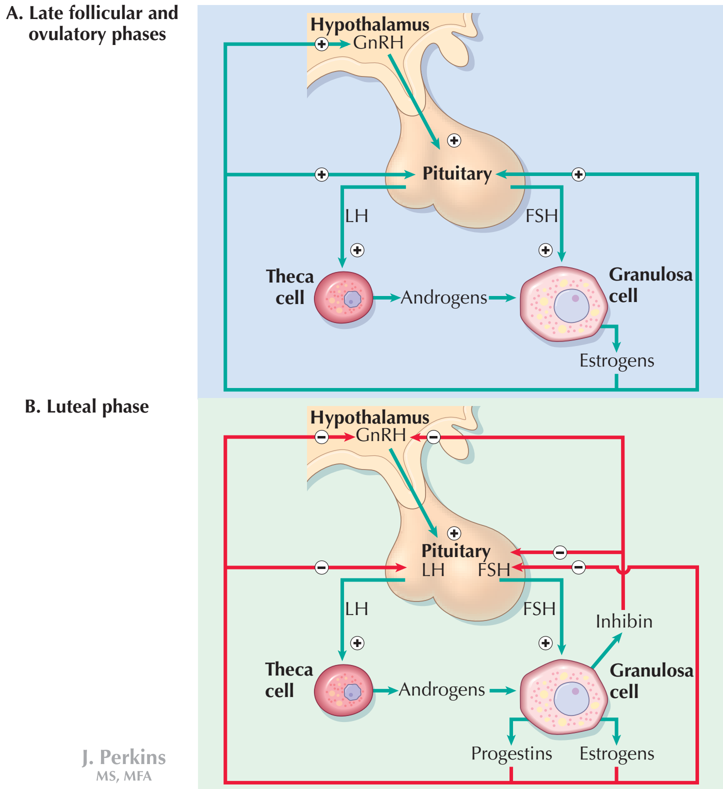


Figure 31.5 **Hormonal Regulation of the Menstrual Cycle** The hypothalamic-pituitary-ovarian axis is characterized by both positive and negative feedback over the course of a menstrual cycle. Initially, GnRH stimulates release of LH and FSH by the pituitary; estrogen synthesized by developing ovarian follicles has negative feedback effects on the axis. However, in the late follicular phase *(A),* blood estradiol reaches a high level that initiates positive feedback and a surge in LH and FSH release, provoking ovulation. In the luteal phase, the system is characterized by negative feedback *(B).* Estradiol, progesterone, and inhibin produced by the corpus luteum have negative feedback actions on gonadotropin release. FSH, follicle-stimulating hormone; GnRH, gonadotropin-releasing hormone; LH, luteinizing hormone.

IMPLANTATION AND PREGNANCY

As discussed, fertilization of the ovum takes place within the fallopian tube. Once the sperm penetrates the ovum, the second meiotic division is completed (all oocytes are arrested in the first meiotic division until a woman reaches puberty; the "mature ovum" released during each cycle has undergone further development and is arrested in the second meiotic division until sperm penetration). Subsequently, mitotic division of the fertilized egg begins and a blastocyst is formed. About 5 days after ovulation, the blastocyst begins to implant in the endometrial lining of the uterus. The sequence of events from follicular maturation to implantation of the blastocyst is illustrated in Figure 31.6.

The **placenta** is formed from **trophoblast** cells of the blastocyst and **decidual cells** of the endometrial lining. **Trophoblasts** secrete a hormone known as **human chorionic gonadotropin** (HCG), which has LH-like actions. HCG "rescues" the corpus luteum from the regression that would otherwise occur at the end of the menstrual cycle, such that synthesis of estradiol and progesterone continues, and thus, the proliferated, secretory state of the endometrium is maintained to support the pregnancy. HCG is important in this regard for the first trimester of pregnancy. By the second trimester, the placenta itself secretes large quantities of **progesterone** and **estrogens** (primarily **estriol**) to support the uterus.

CLINICAL CORRELATE
Measurement of HCG

Most pregnancy tests, including the widely available self-administered tests, are based on measurement of HCG. Although blood tests are more sensitive and may detect pregnancy earlier, urine tests are capable of detecting HCG within a few days of implantation of the blastocyst. Serial measurement of HCG also can be useful in monitoring early progress of a pregnancy when risk of ectopic pregnancy or miscarriage is an issue, because HCG normally rises rapidly following successful implantation.

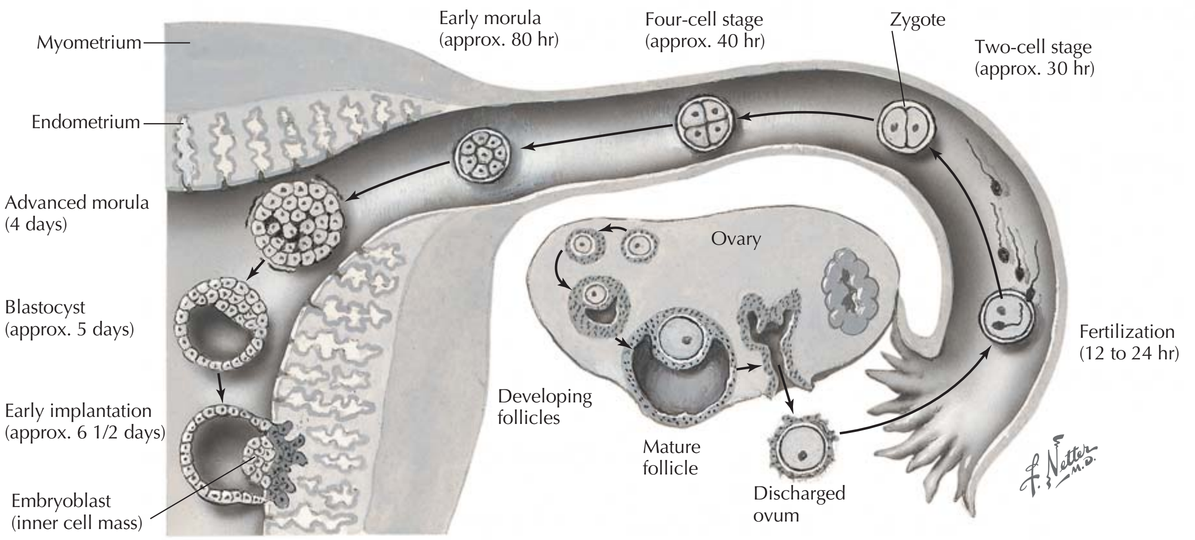


Figure 31.6 **Fertilization and Implantation** Upon rupture of the graafian follicle, the ovum enters the fallopian tube. If fertilization occurs, it takes place within the fallopian tube, which transports the ovum or zygote to the uterus. A zygote will have reached the blastocyst stage by day 5 and will implant in the endometrial lining at that time.

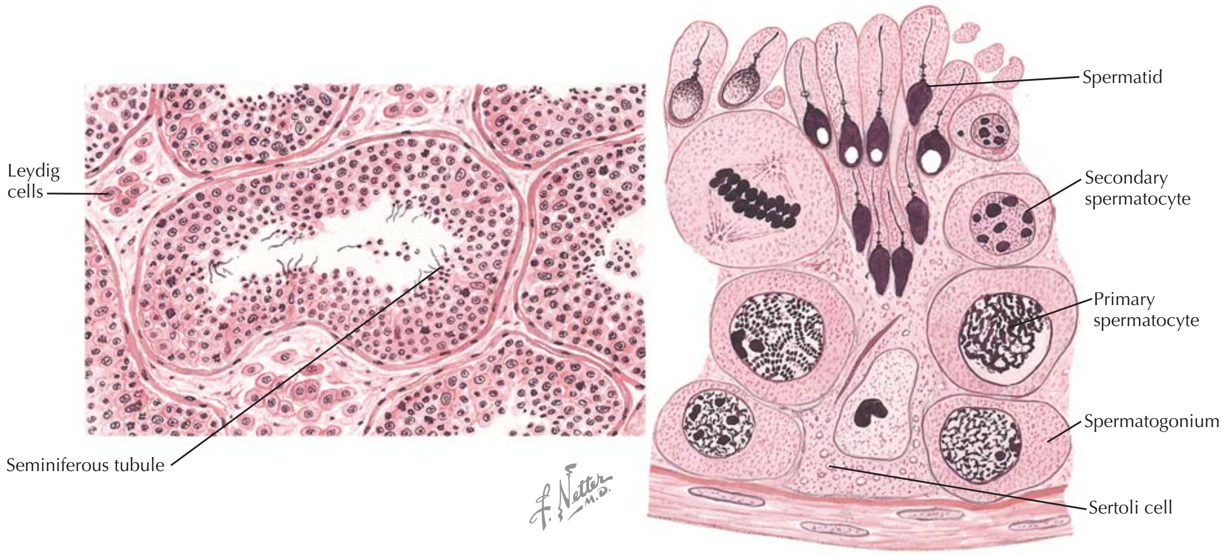


Figure 31.7 **The Testes and Spermatogenesis** The testes contain convoluted seminiferous tubules, where spermatogenesis takes place, and, between the tubules, Leydig cells, which synthesize testosterone in the mature male *(left panel).* Sertoli cells constitute the epithelium of the tubules *(right panel).* Differentiation of primary spermatocytes to sperm cells begins between the Sertoli cells and is completed in the epididymis.

MALE REPRODUCTIVE ENDOCRINOLOGY

The testes are the site of gametogenesis and steroidogenesis in males. Spermatogenesis takes place within the convoluted seminiferous tubules of the testes, whereas the **Leydig cells** located between tubules produce testosterone (Fig. 31.7). The hormone **inhibin** is synthesized and released by Sertoli cells in the lining of the seminiferous tubules. Sertoli cells provide the structure upon which male germ cells develop and secrete fluid that supports flow of sperm through the seminiferous tubules to the epididymis, where further sperm maturation takes place, and sperm are stored until